Augmented Studio

Mediating architectural communication through Tangible Interfaces and Web-AR

Angela Darina Menchise

Master's Thesis
ASEM01- Master Degree Project in Digital Architecture and Emergent Futures

LTH - Faculty of Engineering
Lund University, Department of Architecture and Built Environment

May 2026

Augmented Studio:
Mediating architectural communication through Tangible Interfaces and Web-AR

Author: **Angela Darina Menchise**

Supervisor: **Pablo Miranda**

Examiner: **David Andréen**

Abstract

This thesis investigates the potential of hybrid interaction systems to support architectural communication through the integration of tangible interaction and browser-based augmented reality. Motivated by the communication challenges created by increasingly complex digital design processes, the research explores how physical and digital environments can be combined to facilitate participation, understanding, and collaborative exploration.

Using a Research through Design methodology, the project developed Augmented Studio, a hybrid framework combining physical tokens, augmented overlays, and behavioural interaction modules. Through iterative prototyping, the research shifted from a focus on representation towards an emphasis on interaction, revealing the importance of tangible engagement and relational behaviours within participatory contexts.

The research contributes a framework that combines existing open-source technologies, design methods, and theoretical concepts into an accessible system for architectural communication. The findings suggest that the value of hybrid interaction systems lies not only in visualisation, but also in their capacity to structure dialogue, negotiation, and shared understanding around spatial proposals.

Table of contents

1.Introduction | 7

2.Literature Review | 11
2.1 Complexity and Digital Transformation in the AEC Industry
2.2 Hybrid and Immersive Design Environments
2.3 The Changing Role of the Architect in XR Workflows
2.4 Accessible and Networked Participatory Systems

3.Theoretical Framework | 17
3.1 Social Theory: Participation as a site for intervention
3.2 Cognitive Aspects: Affordances and Hybrid Interaction
3.3 Tangible Interaction and Hybrid Interfaces
3.4 Representation and Meaning: A Semiotic Perspective

4.Methodology | 25
4.1 Research Through Design
4.2 Choice of Technology: Augmented Reality
4.3 Marker-Based and Tangible Interaction
4.4 Web-Based and Platform-Agnostic Approach
4.5 Iterative Prototyping

5.System Development | 33
5.1 Establishing the Web AR workflow
5.2 Spatial Calibration and Tracking Stability
5.3 Development of hybrid system
5.4 Refinement and System Integration
5.5 Behavioural Modules

6. Evidence from the future | 75
6.1 Participatory redevelopment scenarios
6.2 Educational environments
6.3 Distributed architectural practice

7. Discussion | 85
7.1 Reflections on prototype development
7.2 From refinement to interaction
7.3 Participation as mediation
7.4 Contribution through integration
7.5 Future Development and Open Questions

8. Conclusion | 95

9. Acknowledgements | 99

Bibliography | 101

1. Introduction

Architecture depends on communication. Before constructing, visiting or moving in, buildings must be discussed, negotiated, visualised, and understood by a heterogenous group of actors. But, as the tools used in design become increasingly sophisticated, the processes though which architectural knowledge is communicated become more complex and less accessible.

Contemporary architectural practice is driven by an expanding infrastructure of digital technologies. Parameters, Building Information Modelling (BIM), simulations, collaborative platform and computational workflows allow for analysis and production of architecture, but become black boxed to the broader public understanding. This condition becomes particularly relevant in contexts where architectural is expected to engage with multiple stakeholders. Once again, participatory design practices are stating the importance of involving non-professionals and end-users in the discussions about architecture, landscape and urban planning. However, successful participation depends not only on the willingness to engage, but also on the availability of adequate tools able to support communication across different forms of knowledge. When spatial information remains difficult to access or interpret, participation is limited to consultation rather then being active exploration and dialogue.

Recent development in Extended reality (XR) technologies have opened new opportunities for addressing these challenges. Virtual, augmented, and mixed reality offer immersive and situated experiences that can complement more conventional methods. Within architecture, XR is being deployed for

visualisation, simulation, education and collaboration. However, the use is limited by the need of specialised hardware and proprietary software. Thus, the potential of XR technologies for broader participation and communication remains only partially realised.

At the same time, research on tangible interaction underlines the importance of physical engagement in the construction of understanding. Manipulation, gesture, and embodied action provide alternative ways of interacting with information and have been at the base of participatory design methods. Tangible interfaces can create a powerful connection between abstract digital content and physical action, so that a user can engage with complex data through simple interaction. Combining tactile feedback and augmented reality presents an opportunity for researching and reshaping the way we communicate and share knowledge in architecture.

This work investigates the potential of a hybrid interaction system as mediator between professional expertise and participation. In this case, digital and physical are not separated. On the contrary, the research explores how their close integration can support communication and collaborative exploration. A special attention is given to browser-based augmented reality (AR) due to its accessibility. The research is driven by the following question:

How can hybrid interaction systems mediate architectural communication through the integration of tangible interaction and browser-based augmented reality?

The question is addressed by adopting a Research through design methodology centred on iterative prototyping. Through successive iteration of design, fabrication, and testing a hybrid interaction framework named *Augmented Studio* was developed. It is the combination of tangible tokens, augmented reality overlays and interaction modules. Theories drawn form participation, affordance, semiotics and tangible interaction informed the process, while testing constantly reshaped the outcomes.

The objective is not to obtain a finished product or a complete and original methodology. Instead, the project works with accessible and open technology to assemble them into a workflow in support of dialogue around architecture. The main contributions are the prototype and the reflection on the role of interaction, mediation and how the architect fits in the picture.

The work is structured starting from theoretical foundation and moves towards technical implementation and critical reflection. Chapters 2 to 4 give the context of the research, the theoretical framework and define the methodology. Chapter 5 documents the iterative development of the Augmented Studio framework. Chapters 6 to 8 speculate on the potential of the developed system, present critical discussion and revisit the research question to conclude.

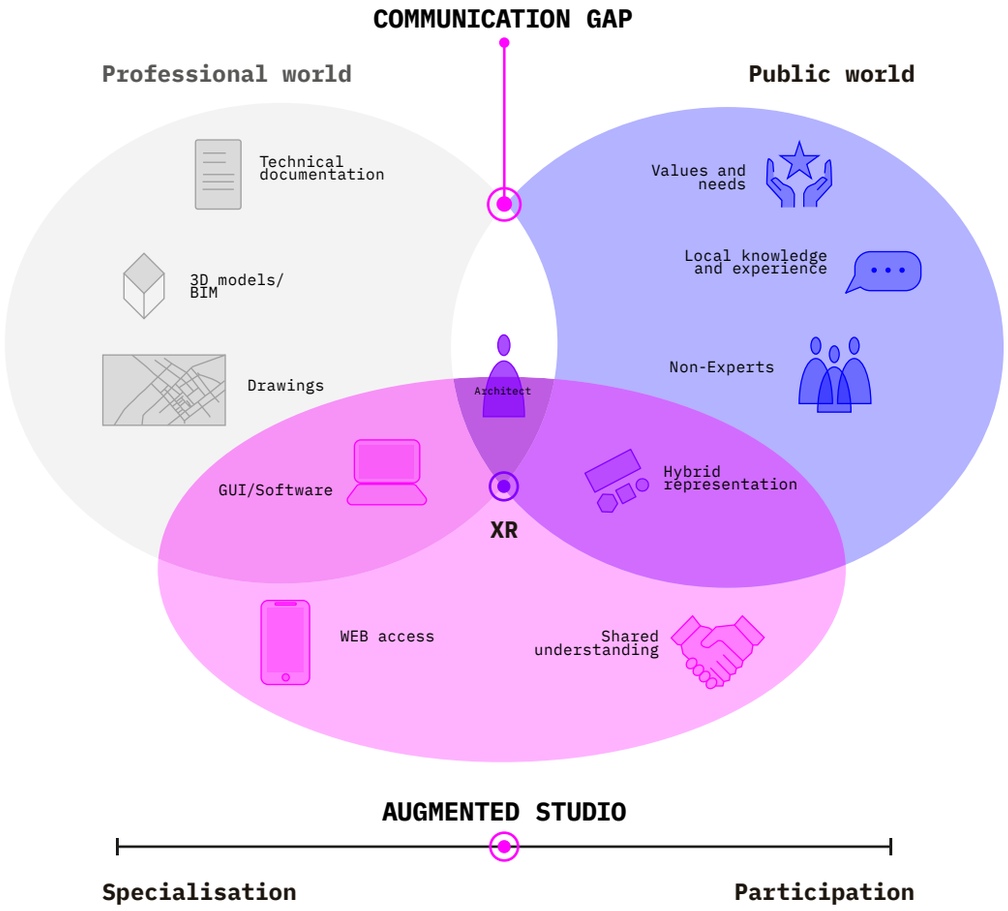


Diagram highlighting the position of the research within the landscape of current practice.

2. Literature review

This chapter examines the relationship between the increasing complexity of the built environment and the parallel development of digital tools within the AEC industry. Through a review of participatory design approaches, XR technologies, and selected case studies, it identifies a persistent gap between technological capability and accessibility in early-stage design processes. The analysed projects highlight both the collaborative potential and the practical limitations of current immersive systems, establishing the conceptual and technological context for the development of the Augmented Studio.

2.1 Complexity and digital transformation in the AEC industry

To understand the potential of a hybrid system for design, the current deployment of Digital Tools - particularly Extended Reality - in the Architecture, Engineering, and Construction (AEC) industry must be examined. The increasing complexity of the built environment has been paralleled by the rapid development of digital tools within the AEC industry. This relationship can be understood as a process of co-evolution, in which advancements in design technologies respond to the growing technical, spatial, and organisational demands of architectural production.

At the same time, these technologies contribute to increasing levels of complexity. While this progression has enabled significant improvements in precision, coordination, and simulation, it has also led to a gradual increase in the opacity

of digital systems. As tools become more sophisticated, they often require specialised knowledge to operate, effectively limiting their accessibility and concentrating their use within expert-driven contexts. As a result, a gap emerges between the technical capacity of these tools and their potential to support broader participation in the design process. This dynamic can be understood in relation to the notion of a “complexity ceiling”, as discussed by John Maeda (2006), where systems that exceed a certain threshold of complexity risk becoming less usable and less accessible. Beyond this point, increased sophistication does not translate into improved performance, but rather into reduced clarity and engagement.

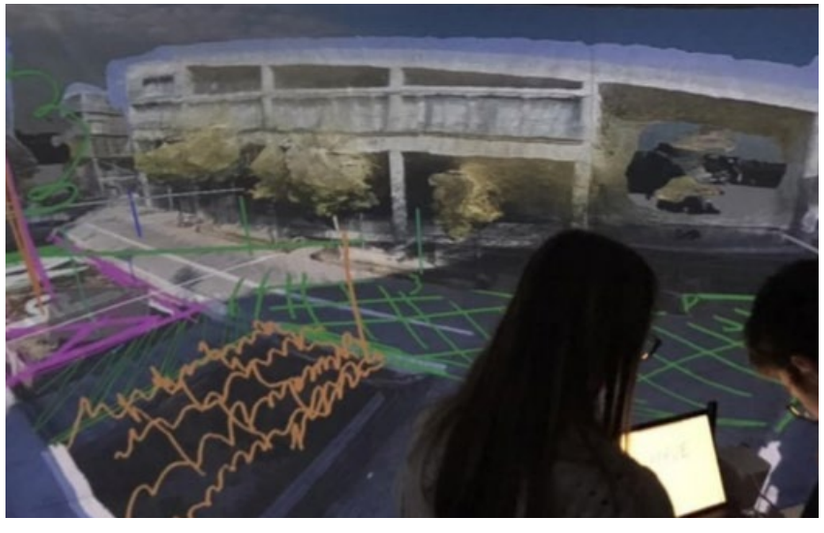
In recent years, Extended Reality (XR) technologies—including Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)—have emerged as powerful tools for spatial visualisation and simulation. These technologies allow users to experience architectural proposals at full scale, enhancing spatial understanding and enabling the visualisation of otherwise abstract or “hidden” aspects of design (Whyte, 2002). As such, XR has demonstrated significant value in design validation, client communication, and on-site applications. Despite these capabilities, a significant gap remains in accessibility, particularly in the early stages of the design process, often referred to as the “fuzzy front end” (Sanders & Stappers, 2008). In these phases—where ideas are still fluid and communication between professionals and non-experts is critical—XR technologies remain underutilised, despite their potential to support shared understanding. Within this context, participatory design practices have long emphasised the importance of involving users as active contributors in the design process.

Foundational work in this field highlights the need to move beyond expert-driven approaches and towards more dialogic and inclusive models of design, where knowledge is shared between professionals and stakeholders (Arnstein, 1969; Blundell-Jones et al., 2012). Traditional participatory methods remain widely used due to their accessibility and immediacy. However, they also present limitations, particularly in their ability to communicate complex spatial relationships and in the discontinuity of the outputs they generate. As a result, there is an increasing interest in hybrid approaches that combine the intuitive qualities of analogue tools with the representational power of digital systems.

2.2 Hybrid and immersive design environments

Several recent projects illustrate the potential of such hybrid methodologies. A primary inspiration for the development of a collaborative “Augmented Studio” is the work of Safin and Dorta (2020), who developed HYVE-3D (Hybrid Virtual Environment 3D) at the University of Montreal. The system challenges conventional uses of Virtual Reality in the AEC industry by shifting away from individual head-mounted displays towards a shared immersive environment. Through a 360-degree projection system combined with handheld devices, HYVE-3D creates what the authors define as a “representational ecosystem”—a space in which physical and digital elements coexist and can be manipulated collectively. A key strength of this approach lies in its support for multimodal communication. Workshops involving non-expert participants integrated physical mock-ups—such as foam and cardboard models—with immersive digital representations. This combination enables participants to engage with design proposals through multiple forms of interaction, supporting different cognitive styles and stages of the creative process. The system acknowledges that design is not a linear activity, but one that benefits from fluid transitions between media. For non-expert users, this is transformative. The intuitive manipulation of spatial elements through embodied interaction allows participants to engage meaningfully in the design process, resulting in high-quality outcomes and increased creative agency.

Hyve-3D in use (Systèmes Hybridlab, 2021).



However, despite its strengths, HYVE-3D presents significant logistical limitations. The reliance on specialised hardware and controlled environments creates a high barrier to entry, reflecting a broader pattern in which increasing technological sophistication can limit accessibility rather than expand it. While the system demonstrates the value of hybrid representational ecosystems, its implementation remains constrained to specific contexts. This thesis seeks to extract the underlying social and cognitive principles of such systems and reframe them within a low-friction, web-based approach. By leveraging widely available devices, the aim is to extend these benefits into more accessible and adaptable environments.

2.3 The changing role of the architect in XR workflows

The emergence of hybrid design environments also implies a transformation in the role of the architect. Rather than acting solely as authors of form, designers increasingly operate as mediators of technologically enhanced collaborative processes. Research by Söderlund and Evans (2022) exemplifies this shift, exploring the cognitive dimensions of immersive co-design. Their work emphasises the value of low-fidelity prototyping within virtual environments, highlighting how simplified representations can support exploration, iteration, and shared understanding. This perspective reinforces the idea that the effectiveness of immersive tools lies not only in their visual fidelity, but in their capacity to facilitate interaction and communication.

This approach finds validation in professional practice through initiatives such as the Zaha Hadid Virtual Reality Group (ZHVR Group, 2018), which has actively integrated XR technologies into architectural workflows. Projects such as Project Correl demonstrate how immersive environments can function as practical tools for coordination, communication, and design development. Together, these examples suggest that the evolution of the AEC field depends not only on the development of new tools, but on the redefinition of design practice itself. Increasingly, value is placed on the ability to design systems for interaction alongside physical artefacts, positioning the architect as a facilitator of shared experiences rather than a sole creator.

2.4 Accessible and networked participatory systems

At a broader scale, the principles of hybrid design extend into urban contexts through open-access and networked platforms. The Networked Participatory Design initiative by Hybrid Space Lab (Sikiaridi & Vogelaar, 2012) exemplifies this approach by enabling citizens to collaboratively design and test digital representations of their neighbourhoods. By reducing proprietary barriers, such systems expand participation and foster engagement across diverse user groups. However, accessibility in these contexts is not determined solely by availability, but also by the design of interaction. The effectiveness of such platforms depends on their ability to provide intuitive and responsive interfaces that support continuous engagement. Recent research by Hu et al. (2025) addresses this aspect by exploring the use of Mixed Reality (MR) to facilitate iterative design processes through actions such as “undoing” and “redoing.” Their work highlights the importance of low-friction feedback loops, where users can quickly test, modify, and refine ideas. This iterative logic is central to the development of accessible design systems, as it reduces cognitive load and encourages experimentation. Together, these examples emphasise two key requirements for contemporary participatory design tools: technological accessibility and cognitive intuitiveness.

The digital prototype in the initial phase of the co-design session (Söderlund & Evans, 2022).

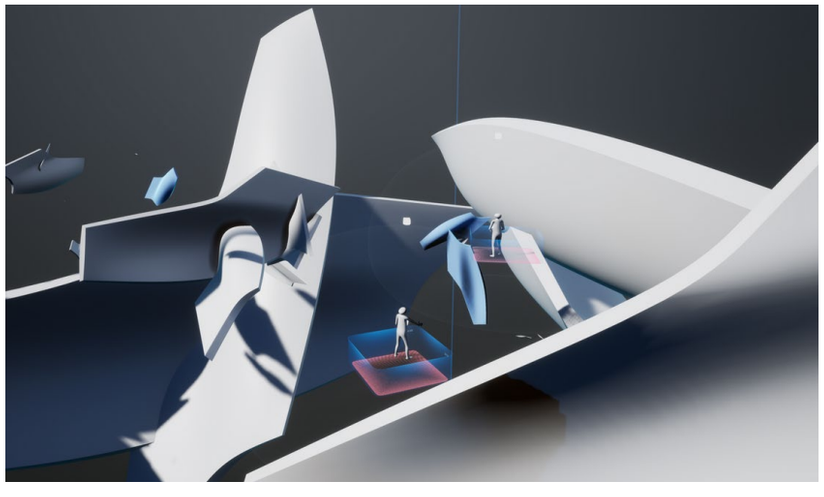


The mediated authors when co-designing on a display box. In the background platonic forms are listed on a white board, beside are colors which can be used to give colour to the platonic forms (Söderlund & Evans,2022).



The examined case studies define a clear trajectory for the evolution of the architectural studio. From the shared “representational ecosystem” of HYVE-3D to the open and scalable platforms explored by Hybrid Space Lab, the value of XR technologies lies in their ability to function as a shared language for communication, rather than solely as tools for visualisation. Despite these advancements, a significant barrier remains in the form of technological complexity and limited accessibility. This reflects a broader co-evolutionary dynamic, in which the increasing sophistication of digital tools risks reinforcing their separation from inclusive and early-stage design practices. By synthesising the multimodal interaction of HYVE-3D, the mediator role identified by Söderlund and Evans, and the iterative logic described by Hu et al., this research identifies a specific opportunity for intervention. The development of a Web-based Augmented Reality system offers a means to reduce these barriers, enabling the integration of immersive tools within participatory design environments. This positions the Augmented Studio as an accessible, hybrid interface capable of supporting communication and co-creation across different levels of expertise. The following chapter establishes the theoretical framework that underpins this approach, drawing on Social Theory, Cognitive Psychology, and Semiotics.

ZHVR Group-Project Correl 1.0 (2018). In Mexico City, the platform ran continuously for three months and attracted over 2,500 creators who collaboratively placed more than 20,000 components.



3. Theoretical framework

This section illustrates the principles that serve as guidelines for the development of the Augmented Studio. Building upon the gap identified in the literature it introduces a set of conceptual perspectives drawn from Social Theory, Cognitive Psychology, and Semiotics.

The motivation for this investigation lies in a persistent tension within the AEC industry between increasing technical capacity and limited design accessibility. As digital tools evolve to address the growing complexity of the built environment, they often become progressively opaque or “black-boxed” (Latour, 1987), restricting their meaningful use to expert contexts. This results in a concentration of technological potential in later stages of the design process, limiting their role in reflective and exploratory phases where communication and shared understanding are critical (Donald Schön, 1983).

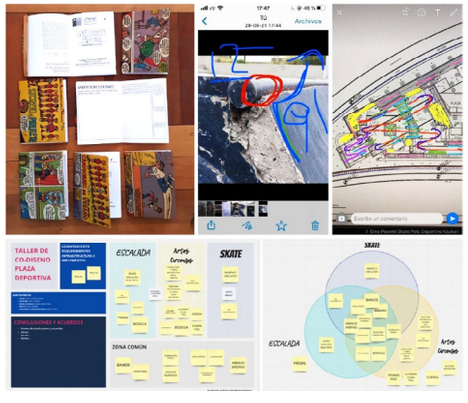
Within this framework, Extended Reality (XR) is examined as a technology with significant potential for improving communication, provided it is integrated through approaches that prioritise accessibility, interaction, and shared understanding.

3.1 Social theory: participation as a site for intervention

Participatory Design Practices (PAD) offer a viable opportunity to loosen the co-evolution loop and bring years of valuable research into the practice field. For the purpose of this research, the participatory environment is not treated as a social end in itself, but rather as a strategic and fertile

“site of intervention”. The Augmented Studio suggests that the conundrum of co-evolution can be addressed within collaborative spaces through hybrid methods of work, combining the social richness of traditional tools with the communicative affordances of immersive technologies.
In these contexts—characterised by a strong emphasis on communication, inclusivity, and low-risk exploration—such systems can be prototyped and put to the test, and subsequently inform other applications. While the complexity problem may require different solutions in other professional and technical situations, this work focuses on the preliminary phases of design.

Historically, Participatory Design has been committed to involve end-users as “partners” in the creation of products and systems. The formalized practice within the built environment, PAD, was driven by the critique of top-down Modernist planning and established the need for designers to prioritise situated expertise over expert opinion (Ehn, 1992). The formalised methodologies relied on tangible, low-fidelity and analogue tools, centred on physically grounded and tactile experiences including charrettes, model making, sketching and public exhibitions (Sanoff, 2000). Although encouraging as means of expression, these tools carry limitations such as challenges in spatial literacy and the discontinuity of gathered data.
From a cognitive perspective, these issues can be related to the difficulty of translating between representational formats and perceived affordances. On the other hand, contemporary PAD practices are transitioning towards hybrid methodologies that include GIS tools, digital whiteboards or non-immersive walkthrough videos to improve spatial understanding (Gaete Cruz et al. 2022). These can be helpful in visualising ideas but lack the non-mediated interactivity of simpler approaches.



A collection of Virtual Collaborative Methods used during the project (Gaete Cruz et al., 2022).

The junction between traditional methods and innovative digital worlds is where the integration of Extender Reality tools becomes beneficial. First, because it restates the importance of PAD practices by bringing them up to date. Secondly, because the aim of this hybridity is not to offer validation, but rather to establish a speculative probe to study the potential of a method that is accessible to anyone, regardless their background.
In this light, collaborative design environments serve as a qualitative laboratory to raise deeper questions. By studying the system through the lens of the non-expert—who represents the highest communication barrier—this research seeks to evaluate the viability of a hybrid “representational ecosystem” (Safin & Dorta, 2020). This sets the stage for an investigation into the cognitive aspects of such a system, exploring the implications of developing a digital-physical interface.

3.2 Cognitive aspects: affordances and hybrid interaction

To understand the potential of hybrid design systems, it is necessary to consider the cognitive mechanisms that underpin human interaction with tools and representations. The concept of affordances, introduced by James J. Gibson (1986), describes the action possibilities that an environment offers to an individual. In the context of design, affordances define how users perceive what actions are possible within a given system. This concept was later adapted by Don Norman (2013), who emphasised the importance of perceived affordances—that is, how clearly these possibilities are communicated to the user.

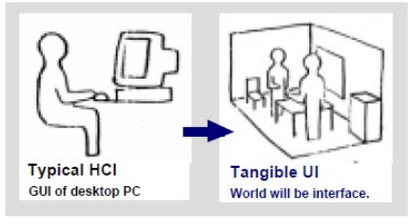
In participatory design contexts, affordances play a critical role in determining accessibility. Traditional tools such as sketches and physical models rely on direct and embodied interaction, allowing users to engage with spatial concepts through intuitive gestures. These forms of interaction require little prior knowledge and support a wide range of cognitive styles. In contrast, many digital design tools introduce layers of abstraction through interfaces that demand technical literacy, creating a barrier for non-expert participants. As a result, the communicative potential of these tools is often limited despite their technical capabilities.

Hybrid systems that combine physical and digital elements offer a means of bridging this cognitive gap. By embedding digital information within familiar physical contexts—such as through Augmented Reality—these systems can align computational processes with intuitive, embodied interaction. This reduces the cognitive effort required to interpret

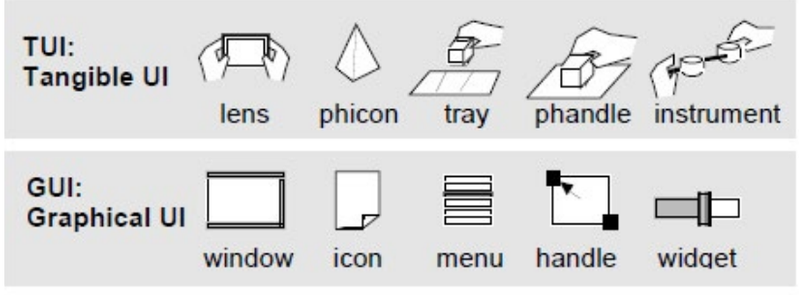
representations and supports a more fluid transition between different modes of thinking and making. In this sense, the effectiveness of a hybrid “Augmented Studio” depends not only on technological performance, but on its capacity to present meaningful, perceivable, and actionable affordances to a diverse range of users.

3.3 Tangible interaction and hybrid interfaces

The integration of physical and digital interaction is further explored through the concept of Tangible User Interfaces (TUI), which seek to bridge the gap between abstract digital systems and embodied human experience. The concept introduced by Hiroshi Ishii and Brygg Ullmer (1997), proposes the integration of digital information within physical interaction, enabling users to interact with digital information through the manipulation of physical objects, grounding computational processes in familiar, material actions. This approach shifts interaction away from screen-based interfaces towards more intuitive and spatial forms of engagement. Subsequent research in Tangible Augmented Reality further explored the integration of physical interaction and augmented environments (Billinghurst, Kato & Poupyrev, 2008). These approaches demonstrate how AR systems can support intuitive manipulation and collaborative interaction by linking digital content to physical objects and gestures.



Diagrams illustrating the new approach to User Intrerfaces and the physical instantiation of GUI elemnts in the TUI (Ishii & Ullmer, 1997).



In the context of participatory design, tangible interaction plays a crucial role in lowering barriers to entry. Physical artefacts such as models, drawings, and prototypes allow users to externalise ideas and engage with design proposals through direct manipulation. These interactions are inherently collaborative and support communication across different levels of expertise. However, traditional tangible tools lack the capacity to dynamically integrate complex data or simulate alternative scenarios in real time.

Hybrid systems that combine tangible interaction with digital augmentation offer a promising direction for addressing these limitations. By linking physical actions to digital responses—such as through Augmented Reality—these systems enable users to interact with computational models in an embodied and accessible way. In this sense, the “Augmented Studio” can be understood as an extension of TUI principles, where physical space becomes an interface for interacting with digital information, supporting both intuitive engagement and enhanced representational capacity.

The role of hybrid systems as mediators between different actors can be further understood through the concept of boundary objects, discussed by Susan Leigh Star and James R. Griesemer (2016). Boundary objects are artefacts that enable communication and coordination across different social worlds, allowing participants with diverse backgrounds and expertise to collaborate without requiring a fully shared understanding.

In design contexts, representations such as models, drawings, and digital environments can function as boundary objects by providing a common reference point for discussion. Hybrid systems that combine physical and digital elements extend this capacity by supporting multiple modes of interaction and interpretation simultaneously. In this sense, the “Augmented Studio” can be understood not only as an interface for interaction, but as a shared representational ground through which professionals and non-experts negotiate meaning and contribute to the design process.

3.4 Representation and meaning: a semiotic perspective

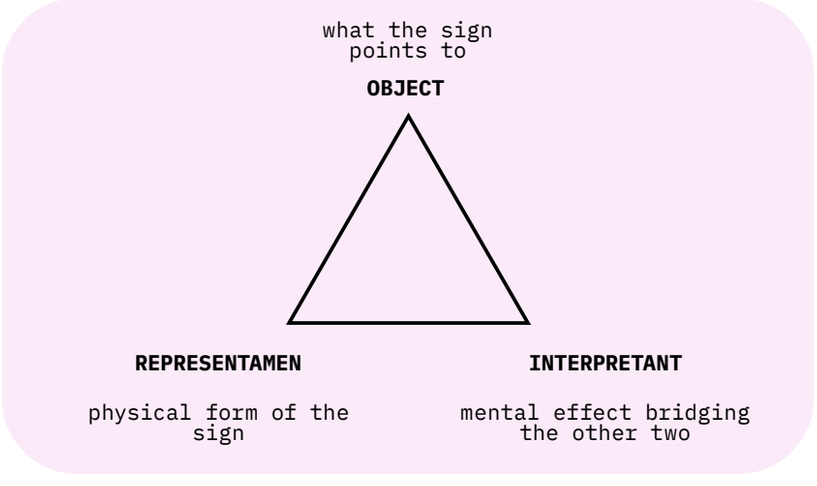
Beyond interaction and cognition, the effectiveness of design tools also depends on how meaning is constructed and communicated through representations. Semiotics, broadly understood as the study of signs and sign systems (Peirce, 1931–1958), provides a framework for understanding how users interpret and assign meaning to visual, material, and spatial

information. In design contexts, drawings, models, symbols, and digital environments function as representational systems that mediate between abstract ideas and shared understanding.

Within architectural practice, representation has historically relied on codified visual languages that require a certain degree of disciplinary literacy to interpret effectively. As discussed in theories of architectural semiotics and representation (Eco, 1980), plans, sections, diagrams, and physical models communicate through conventions that are often intuitive for trained professionals but less accessible to non-expert participants. As a result, participation in design processes frequently depends on translation between different representational modes and levels of abstraction.

From a semiotic perspective, immersive and hybrid environments introduce alternative relationships between sign, representation, and interpretation. Rather than communicating exclusively through symbolic abstraction, technologies such as Augmented Reality situate digital information within embodied and spatially contextualised experiences. In this sense, XR environments can reduce the interpretative distance between representation and perception by allowing users to engage with spatial information through direct interaction and situated visualisation.

Peirce's triadic model of semiotics, showing how meaning is generated through a sign's physical form, its object, and its mental interpretation.

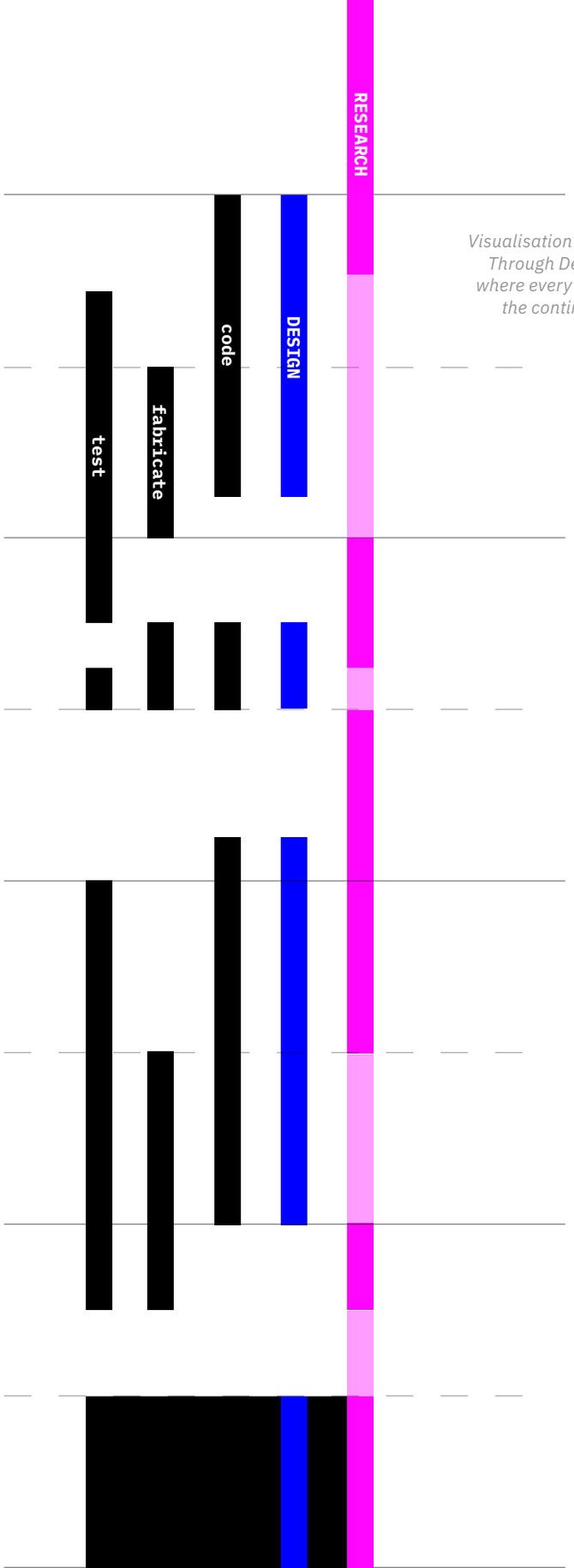


This shift can also be understood through Bolter and Grusin's concept of remediation (2002), where emerging media do not replace previous representational systems but rather reorganise and hybridise them into new forms of interaction. Within Augmented Reality environments, physical models, diagrams, spatial gestures, and digital overlays coexist simultaneously, producing layered forms of representation rather than singular immersive simulations. The medium therefore remains visible as part of the interaction itself, combining physical and digital information into a shared communicative environment.

The development of the "Augmented Studio" can therefore be understood as the construction of a hybrid representational system where meaning emerges through the continuous interaction between physical artefacts, digital overlays, spatial context, and user perception. Rather than functioning solely as a visualisation tool, the system operates as a mediating interface through which spatial ideas can be interpreted, manipulated, and negotiated collaboratively.

The theoretical perspectives outlined in this chapter establish the conceptual foundation for the development of the Augmented Studio. Participation is framed as a strategic context for addressing communication challenges in design, while affordances and cognitive theories highlight the importance of intuitive and accessible interaction. The principles of tangible interaction and hybrid representation further support the integration of physical and digital elements as a means of enhancing shared understanding.

Together, these frameworks inform the design of a system that prioritises accessibility, embodied interaction, and low-friction engagement through Web-based Augmented Reality. The following chapter translates these conceptual principles into a methodological approach, outlining the technological choices and iterative processes through which the system is developed.



Visualisation of the Research Through Design approach, where every phase is part of the continuous research.

4. Methodology

Given the exploratory and practice-based nature of the research, the investigation combines theoretical reflection with iterative prototyping and technological experimentation. The following sections describe the methodological principles that informed the development of the system, including the adoption of a Research Through Design framework, the selection of Web-based Augmented Reality technologies, and the iterative processes through which the prototype evolved.

4.1 Research Through Design

This work adopts a Research Through Design (RtD) methodology, in which design development operates as a reflective and exploratory process (Frayling, 1993; Gaver, 2012). Rather than approaching the Augmented Studio as a finalised product, the system is treated as an evolving framework through which questions of accessibility, participation, and hybrid interaction can be examined.

Given the exploratory nature of the investigation, the methodology combines theoretical reflection with iterative prototyping and experimentation. In this context, the development process also serves to critically examine how immersive technologies can support communication and interaction within early-stage architectural design processes. This perspective aligns with Schön's concept of reflective practice (1983), where understanding emerges through cycles of action and reflection during the design process.

The research therefore prioritises process over optimisation. Rather than seeking a singular or definitive solution, the project focuses on the broader implications of accessible hybrid interaction systems and their potential role within participatory design environments.

4.2 Choice of technology: Augmented Reality

Among the various Extended Reality (XR) technologies, Augmented Reality (AR) was selected due to its capacity to maintain a continuous relationship with the physical environment. Unlike Virtual Reality systems, which often rely on fully immersive head-mounted displays, AR overlays digital information onto existing surroundings, allowing users to remain situated within shared physical space (Abhari et al., 2021).

This characteristic is particularly relevant within collaborative and participatory contexts chosen as sites of intervention, where communication, spatial awareness, and social interaction remain central. The use of handheld AR also reduces some of the social and physical barriers associated with immersive headsets, enabling interaction through devices that are common in everyday life. As smartphones and camera-enabled devices have become widely available, AR offers a comparatively accessible medium through which immersive interaction can be explored.

Within this framework, AR is understood not simply as a visualisation tool, but as a hybrid interface capable of linking digital content to embodied and situated interaction.

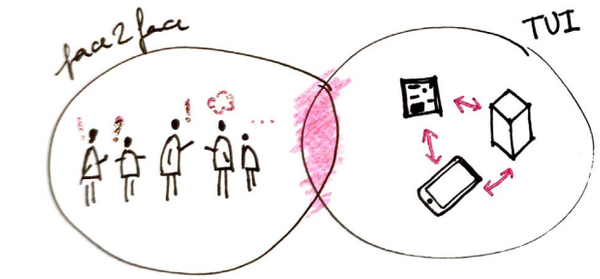
4.3 Marker-based and tangible interaction

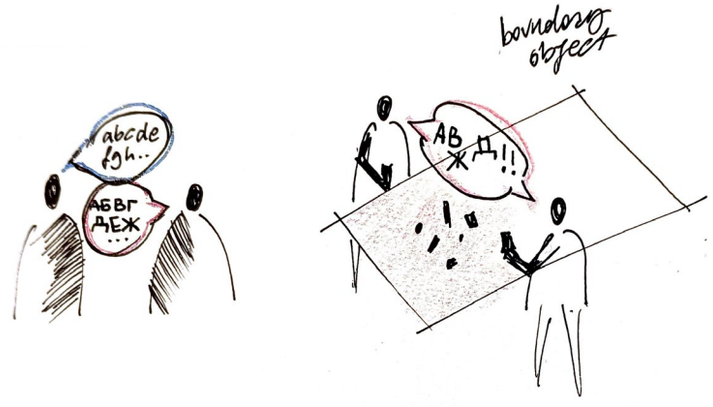
The research adopts a marker-based AR approach as a means of establishing explicit relationships between physical artefacts and digital content. Beyond their technical functionality, markers operate as tangible interfaces that support embodied interaction and reinforce the hybrid logic of the system.

This approach relates to principles established within Tangible User Interface research, where physical objects act as mediators for interacting with digital systems (Ishii & Ullmer, 1997). Similarly, research into Tangible Augmented Reality demonstrates how physical manipulation can support intuitive interaction and collaborative engagement within augmented environments (Billinghurst, Kato & Poupyrev, 2008). Research by Rojtberg and Olwal (2010) further explores the integration of handheld AR systems with tangible interaction, demonstrating the potential of mobile devices as accessible interfaces for spatial and embodied engagement.

Within the development of the Augmented Studio, the marker progressively evolved beyond its purely technical role and was reconceptualised as a “token” — a physical artefact that simultaneously functions as a tracking element, an interaction device, and a tangible mediator between physical and digital environments.

Rather than operating as invisible computational infrastructure, the token externalises interaction through material form and physical manipulation, reinforcing the embodied and participatory logic of the system. In this sense, the token acts not only as a trigger for digital behaviours, but also as a shared object around which interaction, interpretation, and collaboration can emerge.





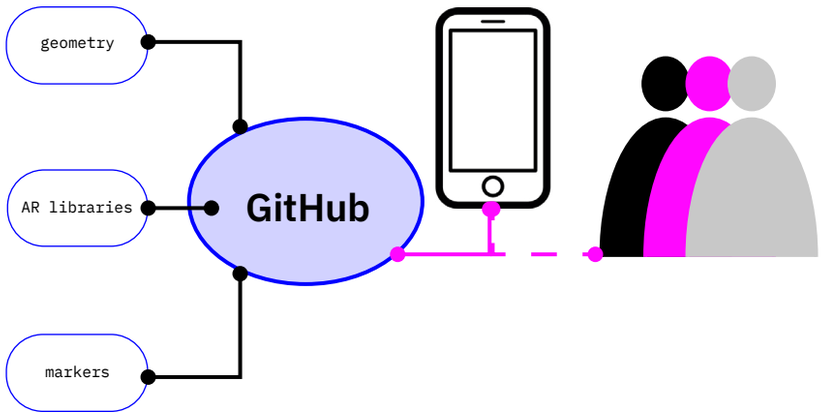
4.4 Web-based and platform-agnostic approach

A web-based and platform-agnostic approach was adopted in order to prioritise accessibility, adaptability, and low-friction deployment. By operating directly through internet browsers, the system avoids dependence on proprietary applications or fixed hardware ecosystems, enabling interaction across a wide range of devices and operating systems.

This approach aligns with recent research emphasising the importance of mobile and hybrid interaction paradigms within Augmented Reality environments, particularly in relation to usability, intuitive interaction, and technological accessibility. Rather than confining immersive experiences to specialised infrastructures, the project explores how AR systems can function within widely available technological contexts and everyday environments.

The use of Web AR also supports rapid iteration and continuous modification throughout the development process. Workflows, interactions, and behaviours can be adjusted without requiring dedicated deployment pipelines or fixed spatial setups, allowing the system to remain flexible and adaptable throughout its evolution.

Diagram depicting the base pipeline of the Web-based system.



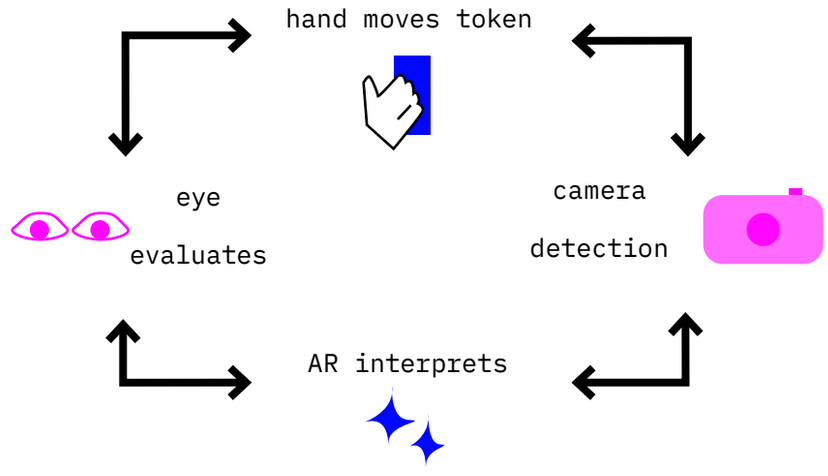
4.5 Iterative prototyping

Iteration forms a central component of the research process. The development of the Augmented Studio evolved through successive cycles of experimentation, testing, reflection, and refinement, where each prototype informed subsequent decisions and adjustments. This iterative logic aligns with Research Through Design methodologies that emphasise experimentation, reflection, and continuous refinement throughout the design process (Gaver, 2012; Redström, 2021).

The iterative process was particularly relevant due to the hybrid nature of the system, which combines digital interaction workflows with physical artefacts and fabricated components. Both the digital environments and the material elements required continuous adaptation in response to usability, interaction quality, and technical feasibility.

This process also extended to the development of the Web AR infrastructure underlying the system. Interaction behaviours, tracking logic, and spatial responses were progressively refined through browser-based scripting workflows, allowing prototypes to be rapidly modified, deployed, and evaluated

Diagram illustrating the iterative feedback loop underlying the development of the Augmented Studio. Each process continuously informs the others, establishing a cyclical interaction process between embodied action and digital feedback.

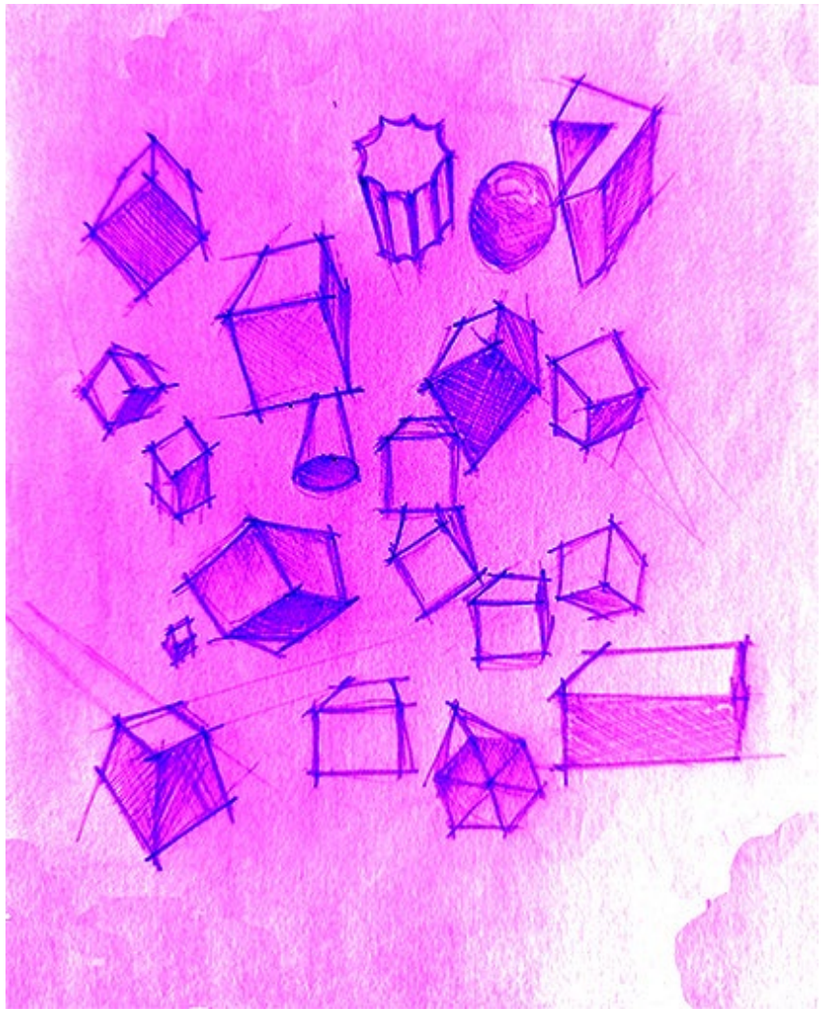


throughout the research process. The use of modular web-based code structures and online repositories supported incremental experimentation and facilitated the continuous evolution of the interactive environments.

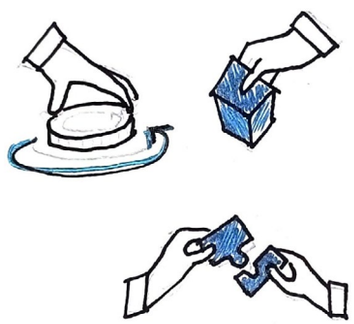
This approach reflects recent XR research emphasising the importance of low-friction feedback loops within hybrid interaction systems (Hu et al., 2025). By allowing interactions to be continuously tested, modified, and reinterpreted, iterative workflows support experimentation while reducing the barriers associated with complex technological systems. The iterative development of the Augmented Studio also relates to recent experiments in collaborative Mixed Reality workflows, such as Reaver's (2022) multiuser participatory design process for the Nordic Pavilion at the Venice Biennale. The project demonstrated how immersive environments can support iterative spatial testing and collaborative decision-making by allowing configurations to be continuously evaluated and modified within hybrid digital-physical contexts.

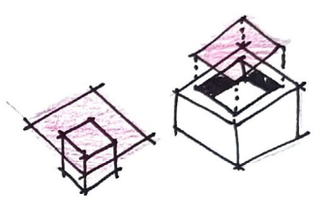
In this sense, prototyping functioned not only as a process of production, but also as a means of refining the relationship between physical and digital interaction within accessible AR environments. Contemporary approaches to tangible augmented interaction, such as Teachable Reality (Monteiro et al, 2023), further reinforce the value of using everyday objects and flexible interaction systems as tools for prototyping hybrid experiences.

Together, these methodological principles establish the operational framework through which the Augmented Studio was developed. The following chapter examines the iterative development of the system, detailing the fabrication processes, interaction behaviours, and technical workflows that emerged throughout the prototyping process.









5. System development and Manufacturing

This chapter documents the development of the Augmented Studio framework through a series of iterative design and prototyping cycles. The chapter presents the workflow, the development of markers and tangible tokens, the implementation of behavioural modules, and the resulting taxonomy of interactions that collectively constitute the final prototype.

The co-evolution discussed in the previous chapters also characterised the development of the Augmented Studio itself. However, in this case, the process was directed towards reducing technological opacity and increasing accessibility. Through successive iterations and continuous adjustments to both code and physical components, the digital and material aspects of the system evolved in parallel, with each process continuously informing the other. This parallel development proved essential, as advancements in the digital infrastructure frequently required the reconsideration of physical artefacts, while material experimentation often revealed new possibilities and limitations for the digital interaction workflows. Rather than following a linear progression, the system developed through interconnected feedback loops between modelling, scripting, fabrication, and spatial testing.

The process began with relatively simple experiments aimed at understanding the available possibilities and constraints of Web-based Augmented Reality workflows. Initial phases focused on establishing functional digital infrastructures and testing different tracking systems, while later stages expanded towards the design and fabrication of physical components and the refinement of interaction behaviours.

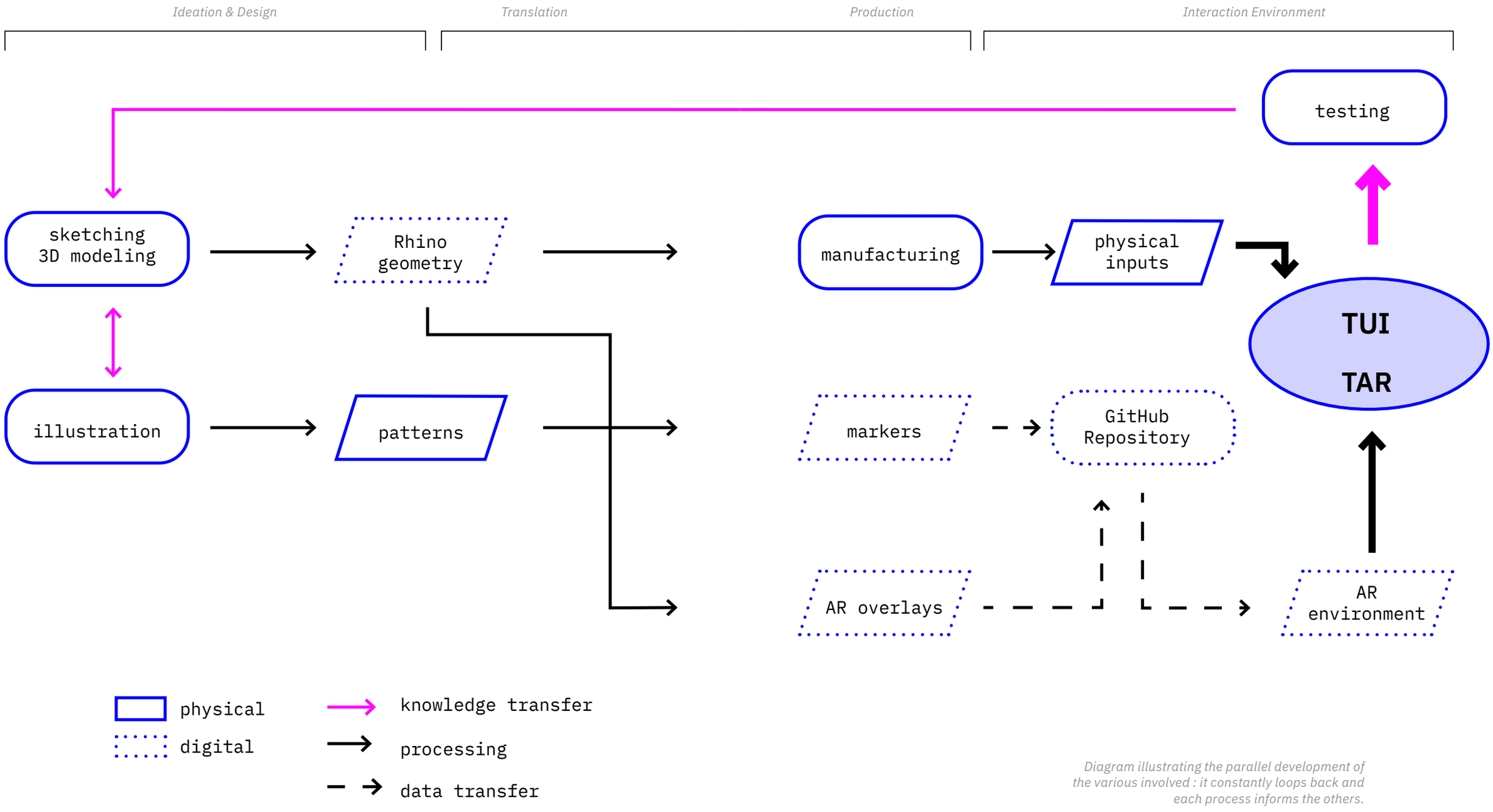


Diagram illustrating the parallel development of the various involved : it constantly loops back and each process informs the others.

5.1 Establishing the Web AR workflow

The first phase of the system development aimed for the structuring of the operational logic for web-based Augmented Reality. The main question was about how the linking of physical and digital happens, and what are the characteristics of the several components involved. All under the guiding principle of it being accessible, non-proprietary and allowing for unlimited runs. Since the researched system requires freedom in what can be done or not, and it is a tool of research, choosing an existing App for AR was not an option, because it both restricts the explorative freedom and often it is proprietary and therefore contradicts the core principle of this work.

The reasonable choice for the preliminary search, fell on Git Hub. Being a cloud-based storage and web hosting manager, it is the most suitable option for combining digital and physical elements into one workflow. The storage is represented by the GitHub Repository, and the web hosting service is taken over by GitHub Pages. To obtain a fully functional, live website URL these two platforms respond to the index file containing instructions in the form of code and pulling external tools to make everything function: a framework to provide a foundational structure and a set of building blocks, and a AR JavaScript (ARjs) library which is a reusable collection of code used to track the camera movements. These open-source tools contribute to the general statement of the thesis about accessibility.

GitHub infrasrtucture

```
[Code: index.html]

    A-Frame draws the 3D + AR.js opens the camera

[GitHub Repository]

    Stores files safely in the cloud

[GitHub Pages]

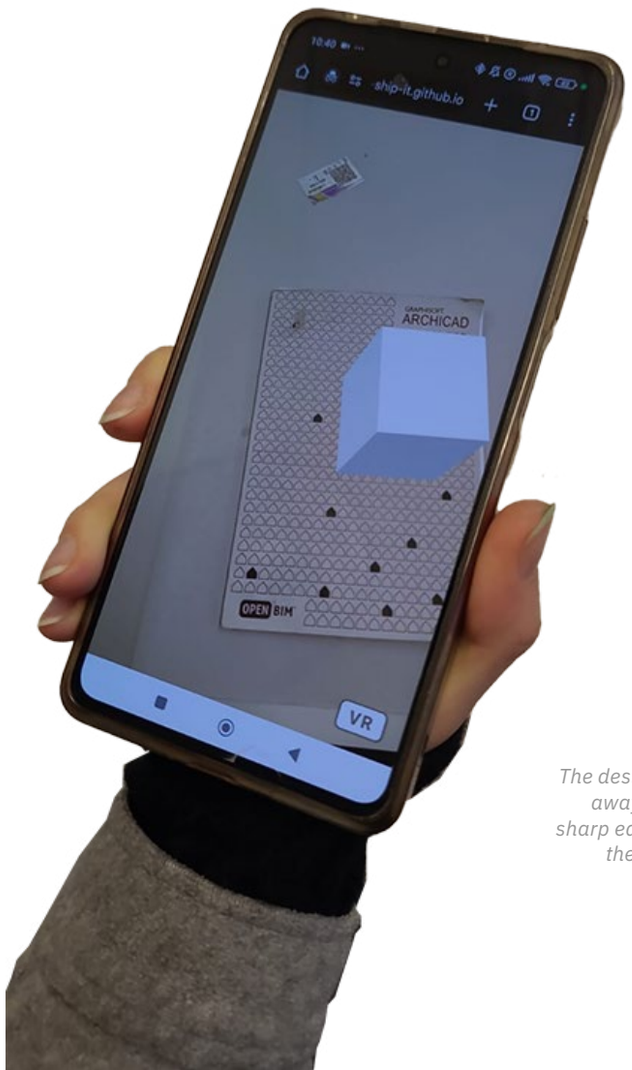
    Turns repository into a live website URL

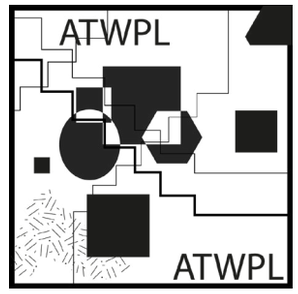
[User's Mobile Browser]

    Opens the camera and renders the AR instantly
```


The Arjs library is specifically designed to bring Augmented reality into the web. It is performative and compatible with most mobile browsers. Of the several tracking methods allowed by the library, image tracking (NFT) was the first to be tested. With the support of A-frame allowing to create 3D shapes quickly within the framework, the pipeline was successfully tested by using the picture of a notebook as marker, processed through MindAR image compiler, to obtain the digital overlay of a cube.

Once the pipeline was established, external geometry in the form of .glb files and personalized marker images could be introduced. However, even the simpler cube experiment already presented issues of tracking stability, scaling, insufficient anchor points, and inconsistency between units and coordinate systems raising key questions for the following iterations. The focus of the parallel work shifted towards the relationship between physical markers, digital overlays and environmental conditions, with calibration becoming a central concern.





The design of a NFT marker. The compiler strips away the colours and analyses the contrast, sharp edges, corners, and unique shapes within the image. It looks for distinct, static spots internalised as feature points.

5.2 Spatial calibration and tracking stability

Introducing .glb files as AR overlays brought up problems in the overlay visualisation. The geometries were modelled in Rhino using centimetres as unit. The image compiler sets the scale to meters for the marker, and this creates a conflict for which a 10-centimetre object would be triggered as a 10 meter one, and thus not visible on a tabletop scale. The issue was solved by introducing a scale factor based of the proportion between the default compiled marker unit (1 meter) and the size of the marker (for the notebook 0.21 meters).

```
<a-entity mindar-image-target="targetIndex: 0">

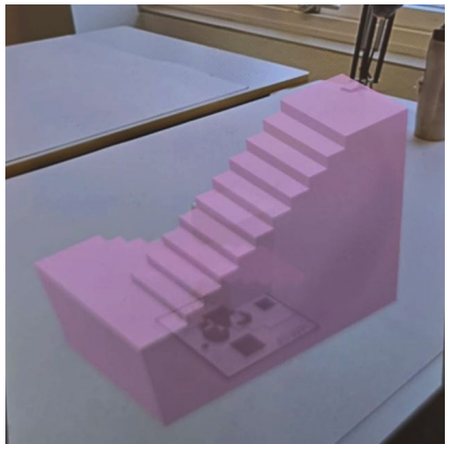
<a-entity position="0 0 -1.44">

<a-gltf-model src="#cubeModel" rotation="0 0 0"
scale="5 5 5"></a-gltf-model>

</a-entity>
```

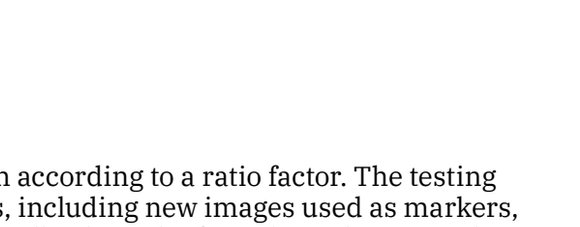
The same component in the code controls the orientation and position of the 3D model. Rhino works with Z axis pointing up, while in ARjs the "up" axis is Y. This should be fixed by the "Map Z to Y" setting in Rhino export window, but some geometry still needed re-orienting through the script, most likely due to mismatching coordinates in the original Rhino file. The translation on the Z axis is by the use of a physical token. A 12-centimetre-tall token requires a shift of the overlay downwards, again according to a ratio factor. The testing of all components, including new images used as markers, provided useful feedback: in the first place, the optimal size for the markers and tokens was defined thanks to the tracking performance.

To take the experiments one step further, the stability of the tracking had to be addressed. Image tracking resulted too sensitive to light conditions and occlusion, and not performative enough when tested with multiple markers and overlays. A more reliable method was tested. An ArUco is high-contrast, synthetic square marker designed specifically for rapid computer vision and 3D spatial tracking. It consists of a solid black outer border that allows a device's camera to instantly detect its shape, surrounding an inner grid of black and white squares (a binary matrix) that represents a unique digital ID number. However, these libraries of markers lack the flexibility required by the investigated digital system, where each component's specific design contributes to the outcome.



The "stair" overlay: if the stair had to be 3 meters, then the rhino model and overlay is 30 cm, so a 1:10 scale. The token is a 1:25 scale average, so it is completely swallowed by the overlay.

Rhino axis system (left) and the ThreeJS axis sytem of the AR enrivonment (right).



10X10cm NFT marker with very stable tracking, but presented problems of occlusion when a token was placed in the central empty space.



A balanced compromise between basic ArUco codes and full image tracking, are ARjs pattern markers. They rely on a strict layout consisting of a thick, solid black outer border—which serve as a highly visible frame for the camera to locate—and a unique, asymmetrical graphic placed cleanly in the centre. Uploading a design into the AR.js online marker training tool, compiles the visual graphic into a lightweight .patt data file, which translates the inner design into a simplified mathematical matrix of pixel brightness values.

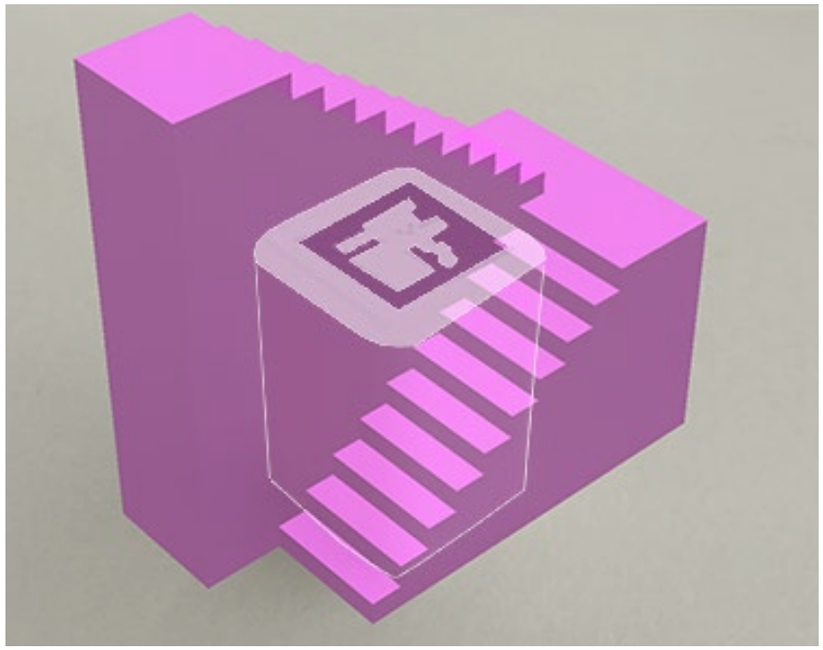
This type of approach is viable for workflows involving multiple markers and their corresponding overlays. Additionally, they improved the tracking stability and glitching of the AR elements detected by the camera. It was therefore kept as core tracking method for the following tests.

```
<a-marker type="pattern" url="./assets/marker.patt">

<a-box position="0 0.5 0" color="red">

</a-box>

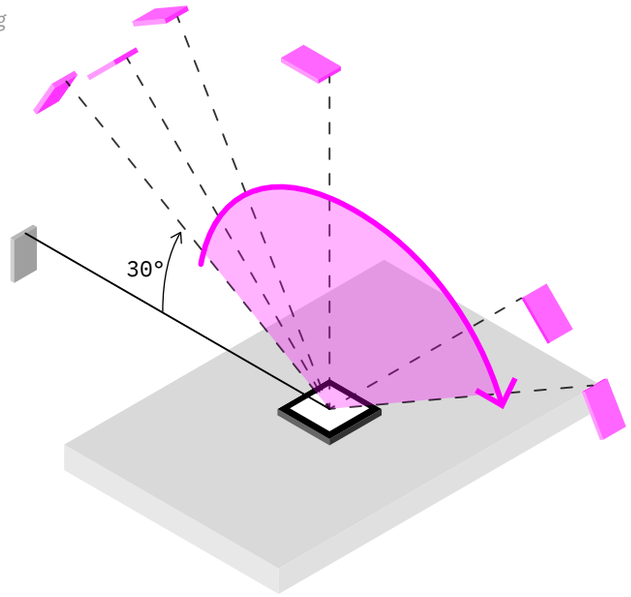
</a-marker>
```

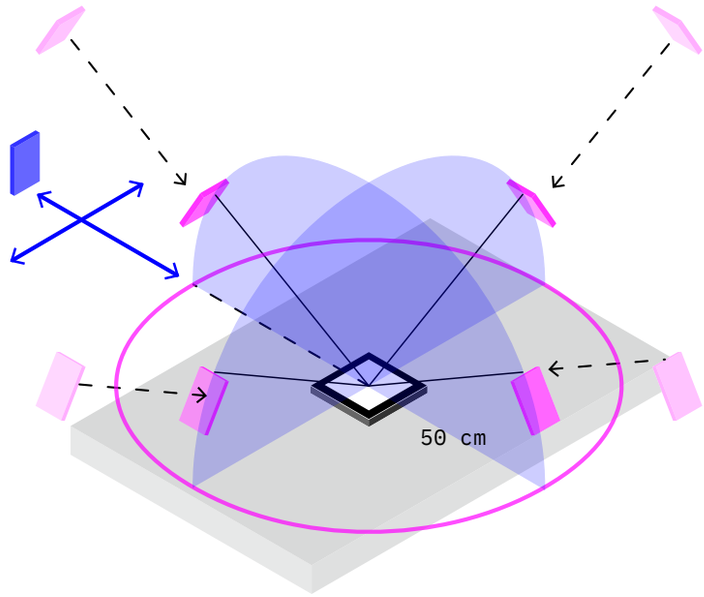


The overlay positioned correctly in relation to a physical token .

The movement of the camera arount the marker allows for a stable result within a 50 cm range over and around the tabletop.

The optimal tracking happens when the camera is angled at minimum 30° .



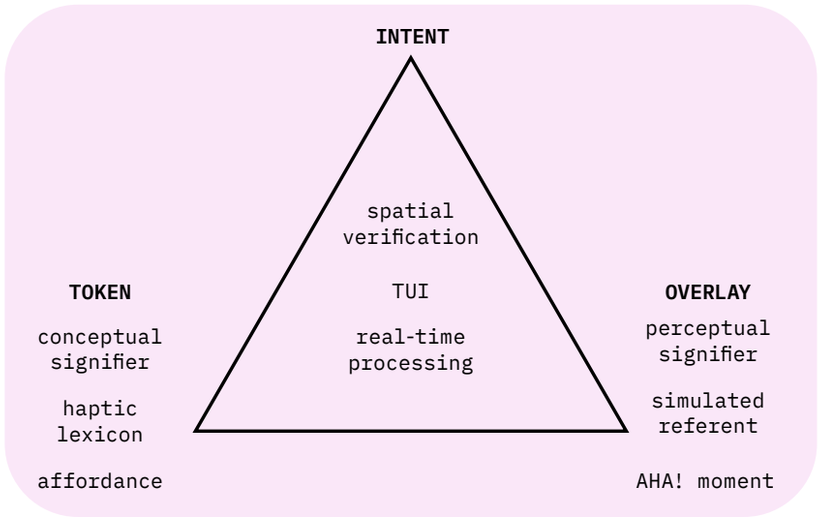


5.3 Development of hybrid system

Once the supporting infrastructure had been established and tested with satisfactory results, the focus of the development process shifted towards the qualitative aspects of the hybrid representational system. During this phase, the physical components gained increasing relevance: the tangible elements progressively evolved beyond their purely technical role and became intentional objects associated with affordance, meaning, and communication. Rather than functioning solely as tracking supports, tokens and markers, began to operate as interconnected layers within a broader interaction framework, where material properties, visual identity, and digital behaviours contributed collectively to the interpretation of the system.

Drawing from the semiotic principles outlined in the theoretical framework—particularly the relationship between signifier, signified, and interpretant described in Peirce's semiotic triad—the hybrid representational system is structured through a conceptual framework connecting architectural intentions, physical tokens, markers, and digital overlays.

Adaptation of Pierce's triadic model.



The matrix of intentions defines the range of spatial qualities and actions that users can express within the hybrid environment. These intentions function as the signified elements of the system, in this case the conceptual meanings associated with boundary, connection, flow, opening, texture, or illumination.

The tangible tokens operate as physical signifiers, translating abstract intentions into manipulable artefacts through form, materiality, and affordance. The Augmented Reality overlays then function as interpretative spatial correspondences, allowing physical manipulations to generate perceptual spatial feedback within the digital layer of the system.

In this sense, the hybrid framework establishes a continuous relationship between conceptual intention, physical manipulation, and spatial visualisation. While the tokens act as conceptual signifiers, the digital overlays operate as perceptual interpretations capable of supporting the evaluation and exploration of spatial intuition.

Conceptual matrix used as guideline for developing the first prototype. Six possible architectural intents are associated with the specific strengh of a token, the type of pattern, and the modality of overlay.

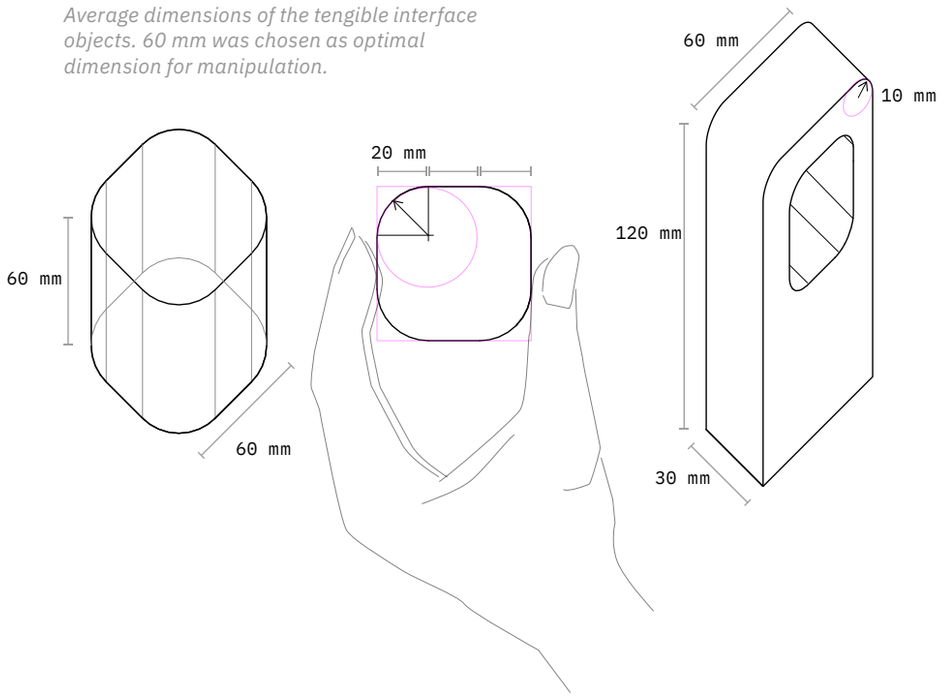
intent	token	pattern	overlay
boundary	shape	index	substitution
connection	interaction	index	substitution
opening	shape	icon	addition
texture	haptic	muted	addition
illumination	shape	index	completion
flow	interaction	index	completion

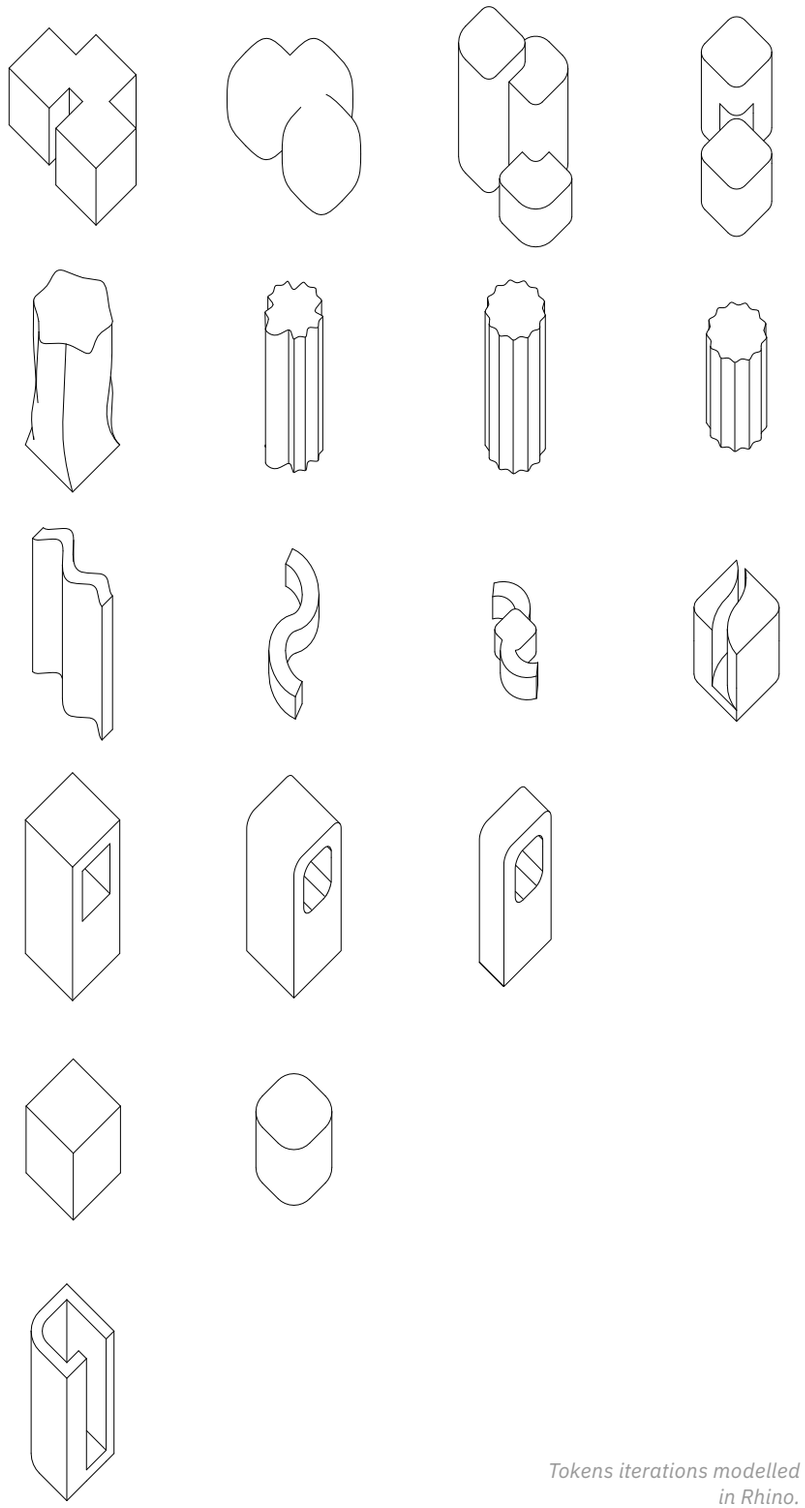
The fabrication of the tangible interfaces combined digital modelling processes with manual craft and material experimentation. Multiple iterations of the tokens were developed and evaluated within the digital environment before being physically manufactured, allowing proportions, affordances, and spatial relationships to be progressively refined.

Material selection was guided not only by fabrication constraints, but also by the intended semantic and tactile qualities of the tokens themselves. Concrete, wood, and PLA were explored in relation to weight, texture, and rigidity while processes such as 3D printing, mould-making, and manual shaping contributed to the physical identity of the system. During fabrication, several interaction-related considerations also emerged, including marker visibility, object stability, ergonomics, and the relationship between token dimensions and digital overlays.

The tokens were developed as objects for direct physical manipulation within shared spatial environments. Their dimensions, mass, surface qualities, and stability influence the ways users handle, position, rotate, and combine them during interaction. In this sense, the system relies on embodied forms of engagement, where spatial exploration emerges through the continuous negotiation between tactile manipulation and augmented visual feedback. The tokens therefore operate as interactive mediators between physical action and digital spatial interpretation.

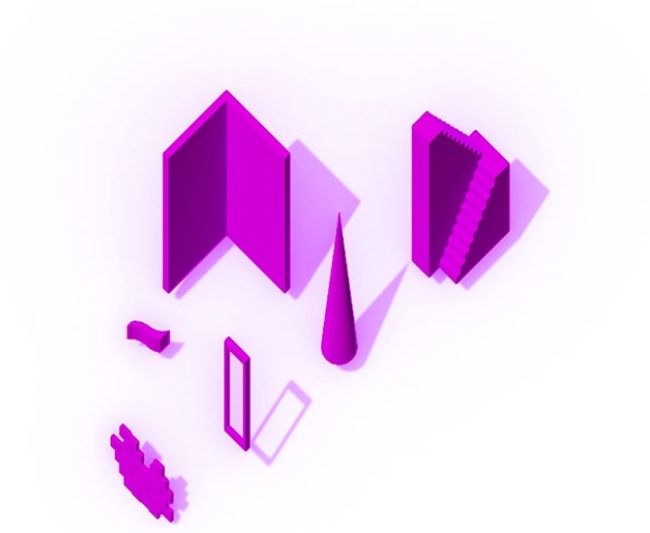
Average dimensions of the tengible interface objects. 60 mm was chosen as optimal dimension for manipulation.





Tokens iterations modelled in Rhino.

While the physical tokens intentionally avoided literal architectural representation in favour of simplified and abstract forms open to interpretation and manipulation, the digital overlays adopted a comparatively more explicit spatial role. In many cases, the overlays functioned as direct spatial interpretations or extensions of the conceptual intentions embedded within the tokens, allowing abstract physical manipulations to generate more legible spatial correspondences within the augmented environment. Three types of overlays were defined in relation to the position the digital model occupies in the dialogue with the physical parts. An addition sees the overlay as a separate entity, while a substitution or completion relation require the digital layer to extend or replace its tangible twin. This aligns to the matrix, which functioned not only as a conceptual diagram, but also as a practical framework guiding the development of the components of the system.



Overlay geometries

The script begins by importing A-Frame and AR.js, which together provide the browser-based 3D and marker-tracking infrastructure for the Web AR environment. A-Frame structures the scene through HTML-like entities, while AR.js enables the camera to recognise printed pattern markers and anchor digital models to them. A custom component, architect-filters, was then defined to control the visual qualities of the imported .glb models, including colour, transparency, and material response. >>

```
<html>

  <head>

    <script src="https://aframe.io/releases/1.2.0/aframe.min.
js"></script>

    <script src="https://raw.githack.com/AR-js-org/AR.js/master/
aframe/build/aframe-ar.js"></script>

    <script>

      // Handles color, transparency, and shading for .glb
models

      AFRAME.registerComponent('architect-filters', {

        schema: {

          color: {type: 'color', default: '#ffffff'},

          opacity: {type: 'number', default: 1.0}

        },

        init: function () {

          this.el.addEventListener('model-loaded', () => {

            const mesh = this.el.getObject3D('mesh');

            if (!mesh) return;

            mesh.traverse((node) => {

              if (node.isMesh) {

                node.material.transparent = true;

                node.material.opacity = this.data.opacity;

                node.material.color.set(this.data.color);

                node.material.roughness = 0.7;

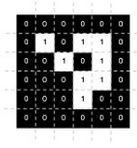
                node.material.metalness = 0.2;

  </script>

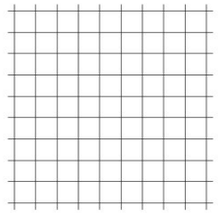
</head>
```

For the design of the markers, the work continued from the spatial calibration and tracking stabilisation phases where the range between 6 and 10 centimetres was identified as suitable for marker-token integration. Moving away from the image tracking and the feature points logic, the pattern markers were designed using ArUco ones as structural references. Namely, the delimited square area of the marker was organized by a 7x7 grid and filled in asymmetrically.

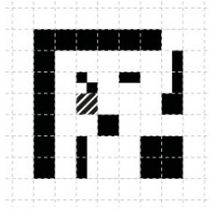
Drawing from Peirce's semiotic classification of signs as icons, indexes, and symbols, the markers operated on multiple representational levels in combination with their physical counterparts. As the system evolved, the visual composition of the markers also began to reflect the conceptual intentions associated with the different interaction categories. Rather than remaining technical tracking patterns, the graphic structures progressively acquired representational and semantic roles within the system and incorporated abstract visual references to the spatial qualities they represented. For example, markers associated with "opening" employed more perforated and discontinuous compositions, while others boundary or connection through their internal organisation. In this sense, the two-dimensional graphic structures complement the three-dimensional qualities of the tokens and overlays, contributing to the legibility and semantic coherence of the hybrid representational system.



ArUco
grid sampling



.patt
tamplate matching



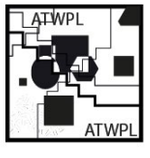
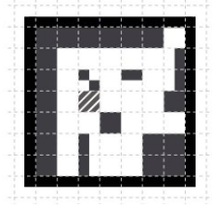
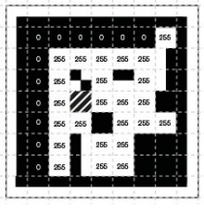


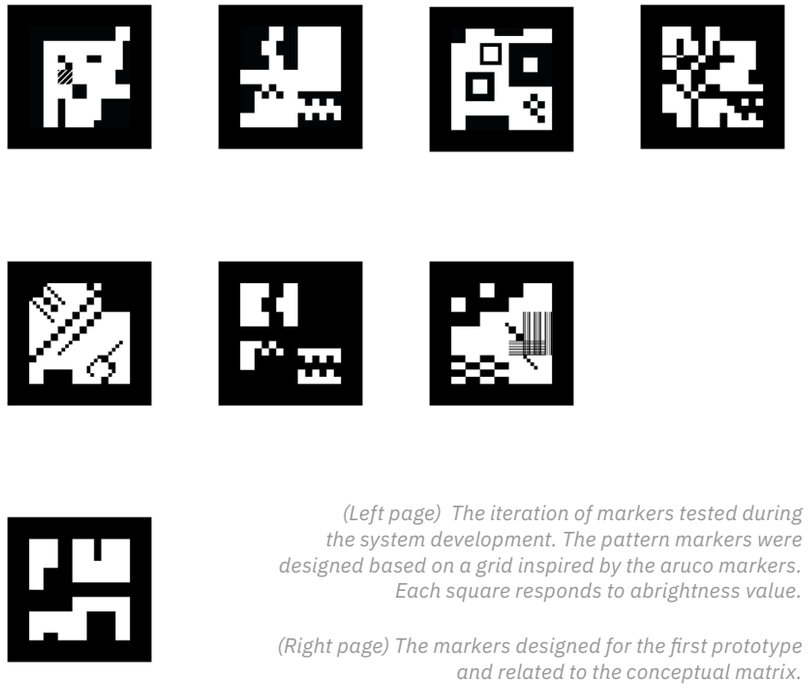
Image tracking
NFT
high contrast





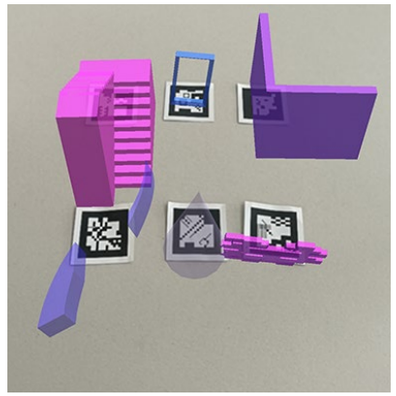
The convergence of the conceptual matrix, the tangible interfaces, the marker system, and the digital overlays led to the emergence of the first complete hybrid prototype. At this stage, the individual experiments conducted throughout the previous phases began to operate collectively as part of a coherent interaction framework, where physical manipulation, visual recognition, and augmented spatial feedback functioned simultaneously.

The first complete prototype provided important observations regarding the operational scale and complexity of the system. The integration of six tokens and corresponding overlays proved sufficiently stable for simultaneous interaction while remaining visually legible and manageable within the shared augmented environment. At this stage, the relationship between marker dimensions, token scale, and overlay behaviour reached a deployable balance capable of supporting interaction without excessive tracking instability or visual overload. The components operated relationally, allowing users to position, rotate, combine, and interpret them within a shared augmented environment.



(Left page) The iteration of markers tested during the system development. The pattern markers were designed based on a grid inspired by the aruco markers. Each square responds to abrightness value.

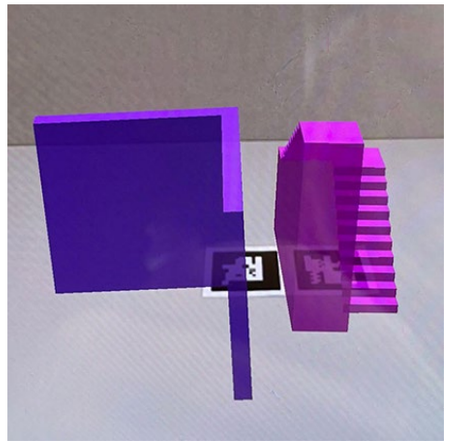
(Right page) The markers designed for the first prototype and related to the conceptual matrix.



<<<

The firts prototype.

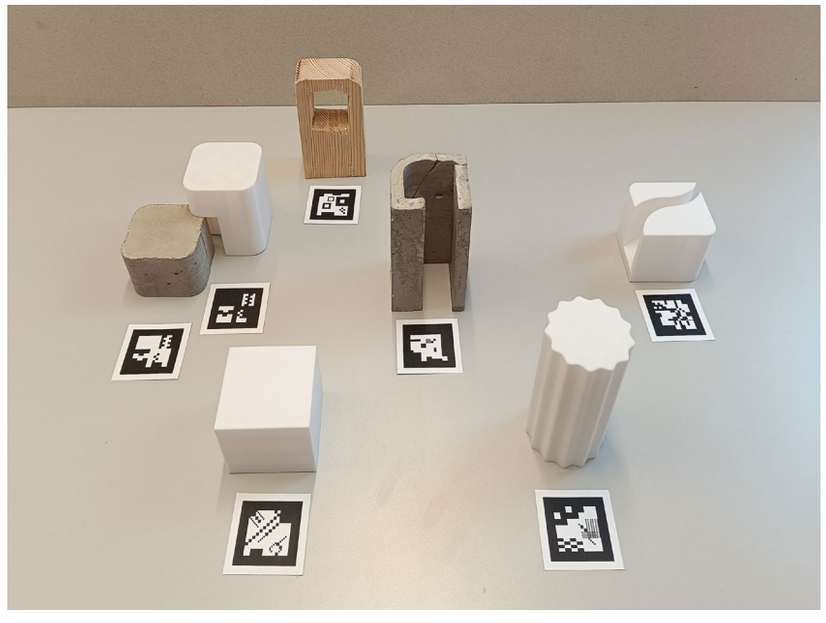
The combination of tangible and augmented elements was explored thorugh arrangement of the parts.

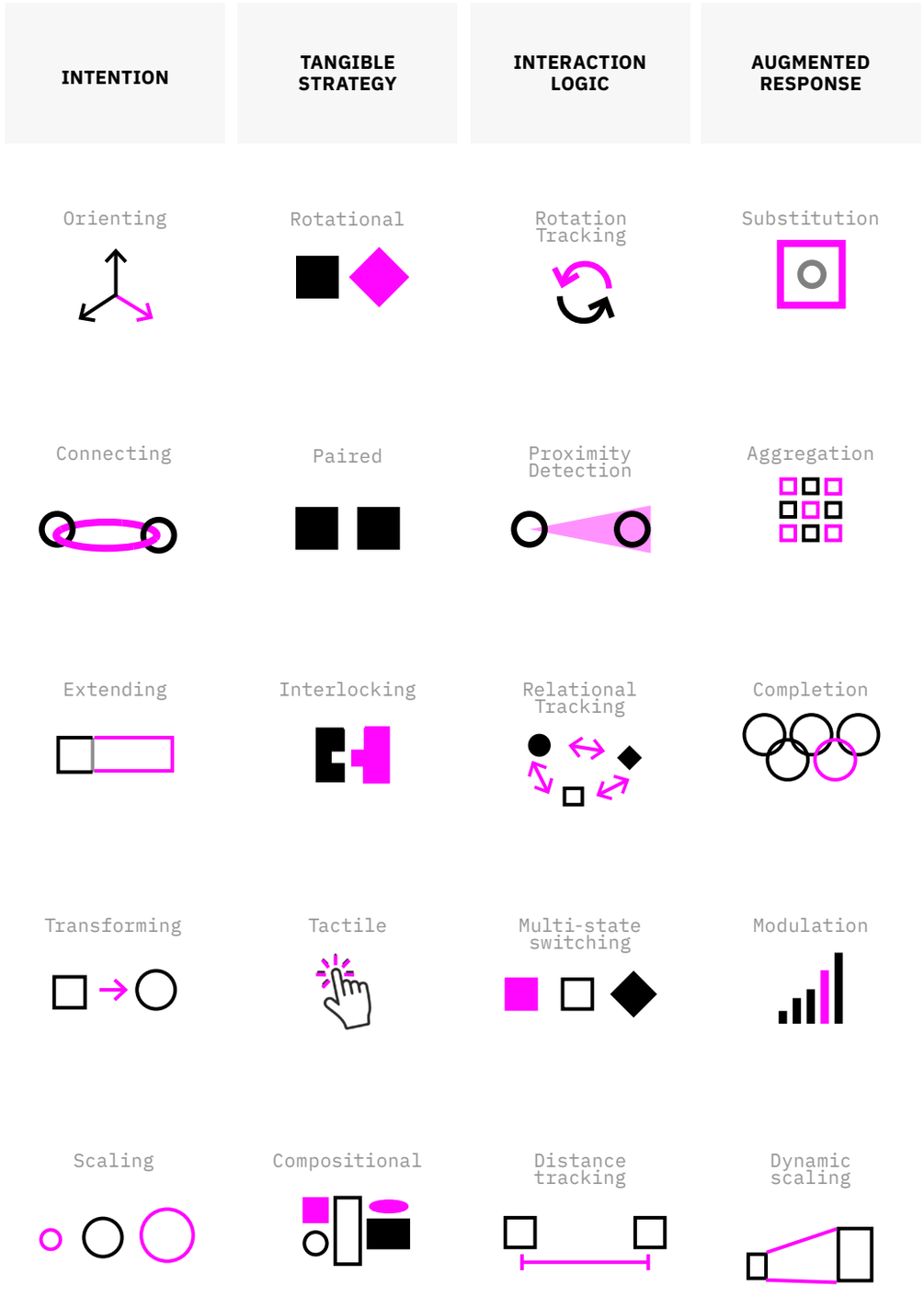


>>>

Taxonomy matrix developed throughout the research.

The framework serves as a guideline for the design, evaluation, and expansion of hybrid interactions by establishing relationships between intentions, tangible actions, digital behaviours, and augmented responses.



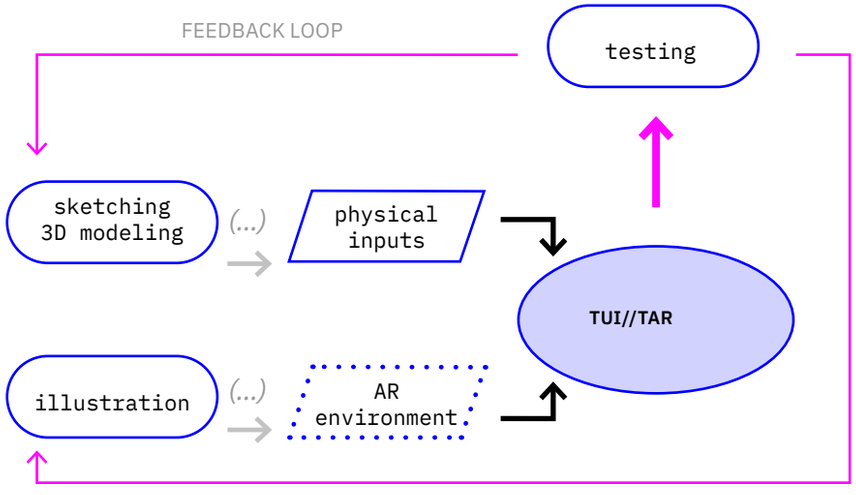


5.4 Refinement and system integration

The first complete prototype demonstrated the viability of the hybrid representational framework at an operational scale, while simultaneously revealing new possibilities and limitations related to interaction, integration, and behavioural complexity. These observations informed a subsequent phase of refinement focused on the relationships between tokens, markers, overlays, and user interaction.

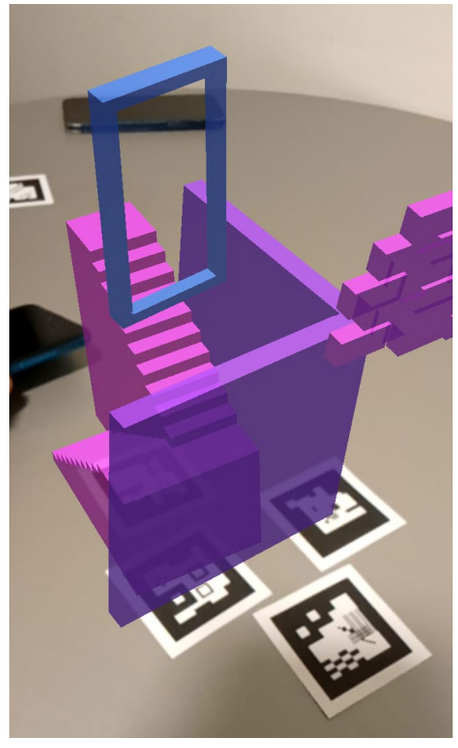
At the completion of the first set, an informal exploratory interaction was conducted with two participants. The goal was to observe how users engaged with the system through manipulation, spatial arrangement, and interpretation of the overlays while using their own camera-enabled devices. Although no specific instructions were provided, the participants navigated the system intuitively and the interaction generated several relevant observations while also raising important questions regarding usability and interaction. During a post-interaction verbal feedback session, the experience was described as engaging and accessible, while the responsiveness and coherence of the digital-physical interface were identified as aspects requiring further refinement.

A summary of the loop illustrating the impportance of testing each iteration and learning from it.



The experiment also revealed several ambiguities: at this stage, the pattern markers existed as detached printed elements rather than integrated components of the physical tokens. While this initially generated inconsistencies within the interaction workflow, it also revealed unexpected modes of use, as participants occasionally manipulated the markers and physical artefacts independently. In some situations, the tokens began to operate as spatial modelling elements while the markers functioned separately as activation devices for the digital overlays. These observations reinforced the importance of establishing greater cohesion between the physical, graphic, and digital layers of the system, leading to a subsequent phase of marker-token integration and fabrication refinement.

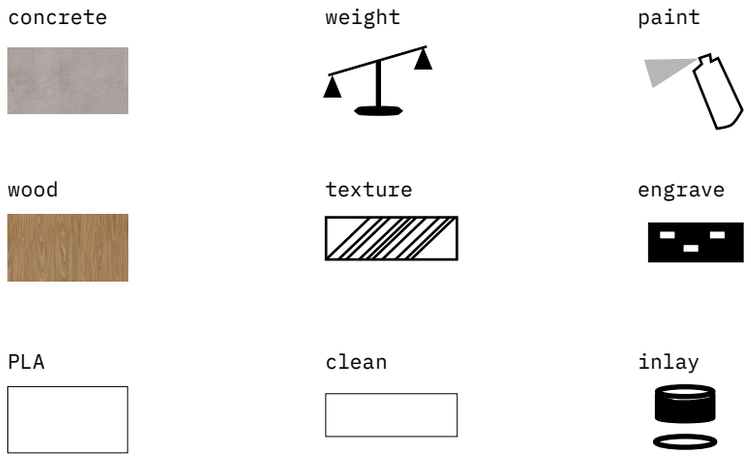
Having chosen three main materials for the manufacturing of the tokens, the strategies for homogeneous integration of the patterns varied accordingly. One common requirement remained the contrast between black and white, as pattern markers rely on the mapping of pixel brightness values and the use of different colours onto already textured surfaces could negatively affect tracking stability.



A screenshot from a user's device during the informal testing of the first prototype. The test showed intuitive aggregation of parts and quick understanding of the system.

As a result, different fabrication approaches were explored in relation to the material properties of each token. Wooden elements were engraved in order to embed the patterns directly into the surface while maintaining tactile continuity, while concrete tokens employed painted applications produced through mirrored stencil techniques to preserve sufficient visual contrast on irregular surfaces. PLA prototypes instead experimented with 3D-printed inlays and filament differentiation as a means of integrating the marker within the object itself. These processes transformed the markers from detached graphic elements into physically embedded components of the tokens, reinforcing the cohesion between material, visual, and digital layers within the system. The resulting integrations proved sufficiently legible for the camera-based tracking system, allowing the markers to remain detectable despite the variations in material, texture, and fabrication method.

Following the integration of the physical and graphic components, the refinement process expanded towards the exploration of more dynamic and relational interaction possibilities, as suggested by the participants in the informal test. Initial experiments focused on the behaviour of the markers themselves, investigating whether individual markers could be fragmented into multiple readable sections and whether spatial relationships between separate markers could be used to trigger overlays conditionally. Proximity-based interactions between markers proved particularly promising, allowing digital content to respond to distance and relative positioning within the augmented environment.



Parallel investigations explored the possibility of introducing relational behaviours between tokens and overlays. Rather than functioning as isolated correspondences, the components began to influence one another through conditional interactions, including overlay transformations linked to marker rotation, multiple overlays associated with a single marker, and cause-effect relationships between different physical entities within the system. Additional experiments also investigated the application of graphic textures and material qualities onto the digital overlays through externally mapped image files.

While several of these experiments remained exploratory, their combined outcomes demonstrated the potential of the system to move beyond static representation towards modular and responsive interaction behaviours. These investigations ultimately informed the development of the five interaction modules discussed in the last part of this chapter.

The refinement of the physical forms progressively shifted from representational concerns towards interaction-oriented geometries capable of supporting relational and behavioural operations within the augmented environment. As the interaction logic of the system evolved, the physical tokens were

(Left page) A chart showing which materials were used for the tokens, their main strenght, and how the markers were integrated

(Below) A piece of the code regulating the overlay trigger depending on the rotation of the marker.

```
const v1 = new THREE.Vector3(0, 0, 1);
v1.applyQuaternion(markerModel.object3D.quaternion);

let angle1 = Math.atan2(v1.x, v1.z) * (180 / Math.PI);
angle1 = (angle1 + 360) % 360;

let activeId;
if (angle1 >= 315 || angle1 < 45) activeId = 'model1';
else if (angle1 < 135) activeId = 'model2';
else if (angle1 < 225) activeId = 'model3';
else activeId = 'model4';

const models = ['model1','model2','model3','model4'];
models.forEach(id => {
    const el = document.querySelector('#' + id);
    if (el) el.object3D.visible = (id === activeId);
});

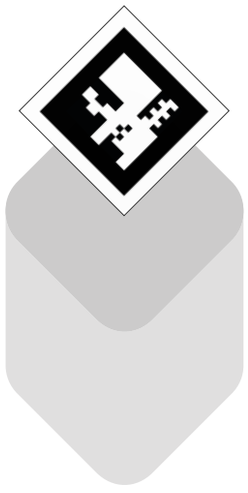
const activeModel = document.querySelector('#' + activeId);
if (!activeModel) return;
```

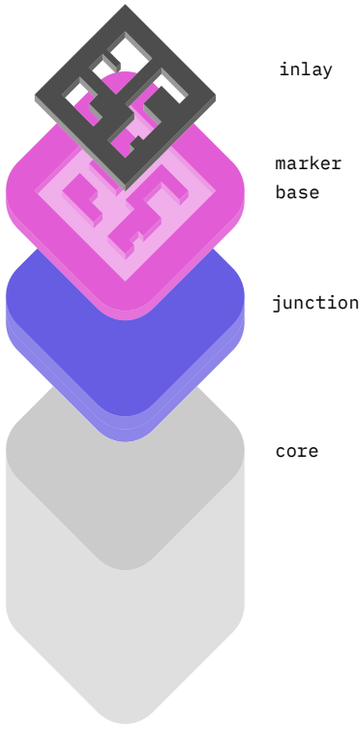
reworked in order to improve handling, tracking stability, and spatial readability during collective use. A more homogeneous dimensional range of approximately 6×6×6 centimetres was progressively adopted, allowing the markers positioned on the upper surfaces of the objects to remain more consistently visible and levelled during interaction. This refinement also proved beneficial for the emerging proximity-based behaviours, where multiple markers needed to remain simultaneously detectable without one prevailing over the other within the camera view.

The introduction of rotational and relational interaction logics also influenced the affordances and morphology of several tokens. Some objects were redesigned with movable components attached to stable cores, enabling users to manipulate orientation-dependent behaviours directly through physical rotation. Other tokens explored interlocking configurations, where multiple physical pieces combined to compose complete markers or activate shared overlays, introducing more explicit relationships between physical assembly and digital response.

(Left page) A token with integrated core and 3D printed marker marker, compared to previous version with paper marker.

(Right page) Second set of digital overlays developed based on the taxonomy.





The refinement of the digital overlays focused primarily on expanding the architectural vocabulary available within the system. Additional overlays were developed to support categories such as boundaries, connections, and transitions, including walls, corners, tunnels, stairs, and openings. Alongside these spatial correspondences, more specific overlays were also produced in relation to the emerging behavioural experiments, including texture variation, state changes, and proximity-triggered interactions. While the distinction between additive, substitutive, and complementary overlays remained conceptually relevant, the later stages of development prioritised relational behaviours and interaction dynamics over the further expansion of representational categories.

Together, these refinements shifted the system from a collection of hybrid representational elements towards a more dynamic and modular interaction framework, laying the foundation for the behavioural structures discussed in the following section.

Switch token



Connection token



Data token



Boundary token



Switch token







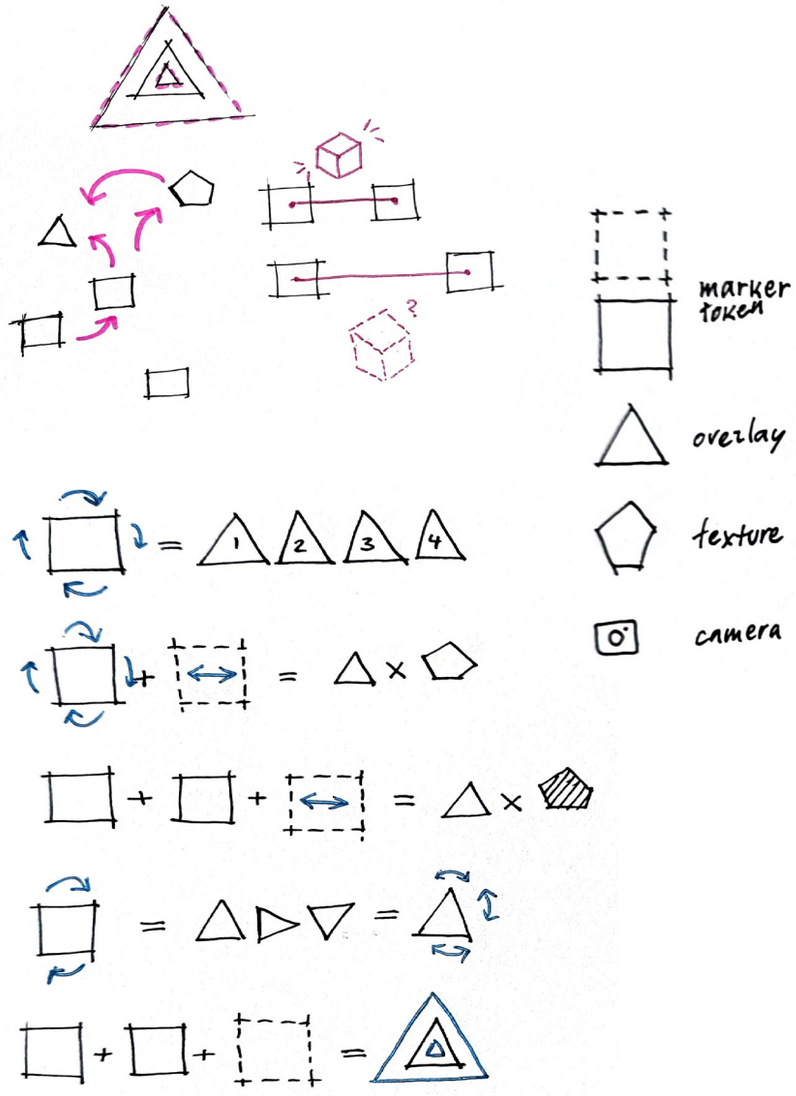












The behavioural modules were originated from sketches, diagrams and notes illustrating what was the expected outcome and deployment. Each module is illustarted with the hand skatches, the tangible toolkit and some relevant piece of code regulating the behaviour.

The repositories, deployments, and supplementary materials associated with the project are publicly available through the author's GitHub profile: https://github.com/an3732me-s-ship-it

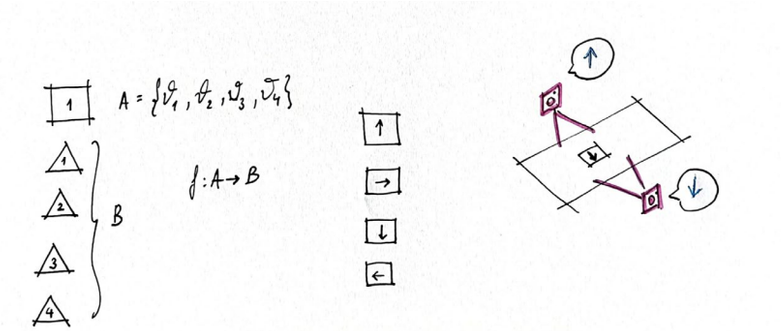
5.5 Behavioural modules

The experiments discussed throughout the previous sections progressively converged into a series of behavioural modules intended to explore different relationships between physical manipulation and augmented spatial response. Instead of functioning as fixed tools or isolated demonstrations, these modules operate as flexible interaction structures that can be combined, adapted and improved within different participatory and representational scenarios. Through the integration of rotational logic, proximity detection, conditional activation, and overlay transformation, the system evolved from a collection of tangible-digital correspondences into a more dynamic and relational hybrid environment. The following modules summarise the primary interaction logics developed within the Augmented Studio framework. They do not represent finalised applications, but exploratory interaction systems through which alternative methods of collaborative and accessible design can be investigated.

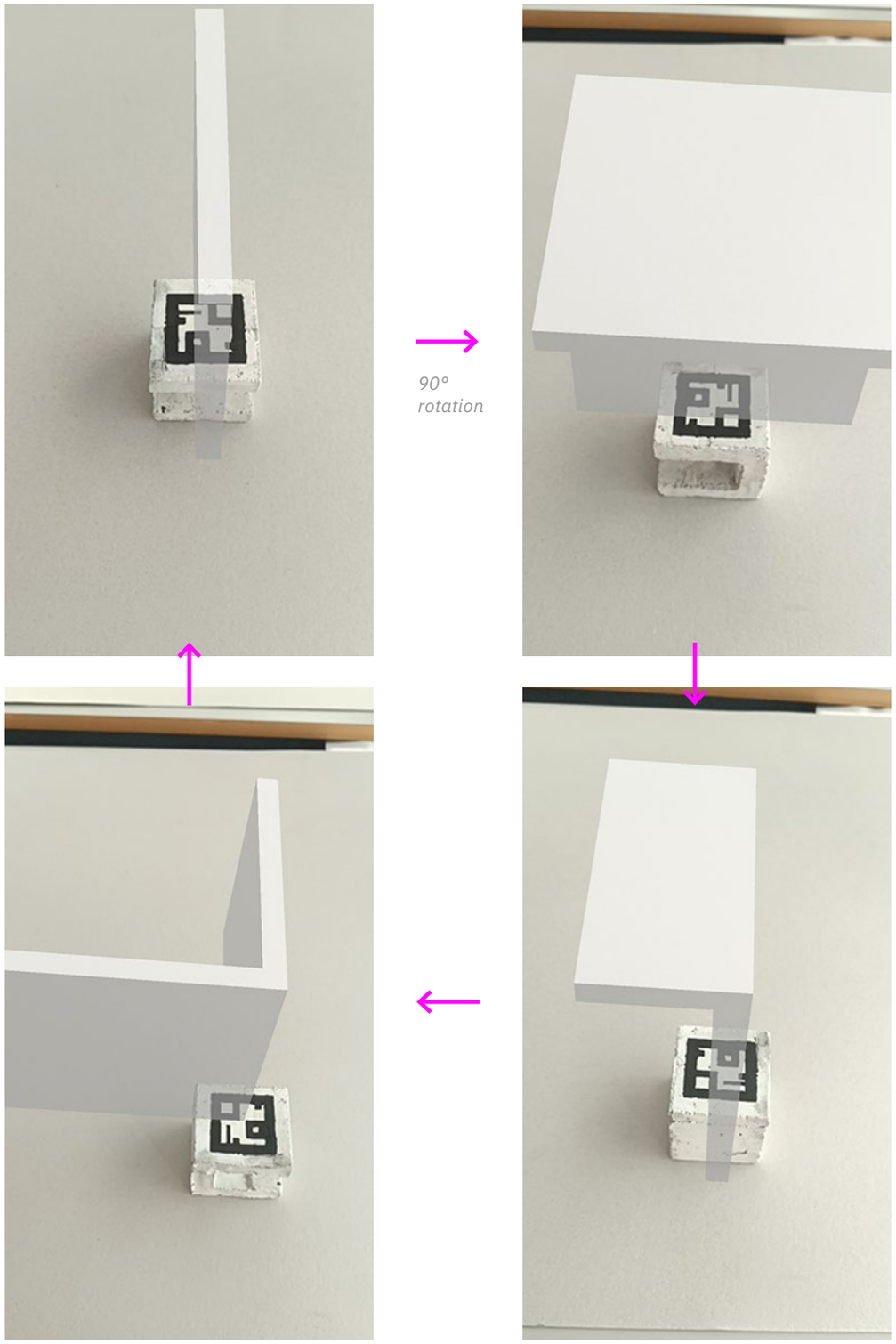


Model Swap

This module explores the possibility of increasing the representational flexibility of the system while reducing the number of required physical components. Unlike the earlier prototype environment where each marker corresponded to a single digital overlay, the Model Swap module allows one marker to trigger multiple overlay variants through rotational interaction. The system associates different orientations of the token with alternative spatial configurations belonging to the same architectural category. Through this mechanism, one physical element can communicate multiple spatial possibilities while maintaining a compact and modular interaction workflow. It is not about a fixed one-to-one correspondence, but about the relationships between parts supporting more open-ended spatial exploration.

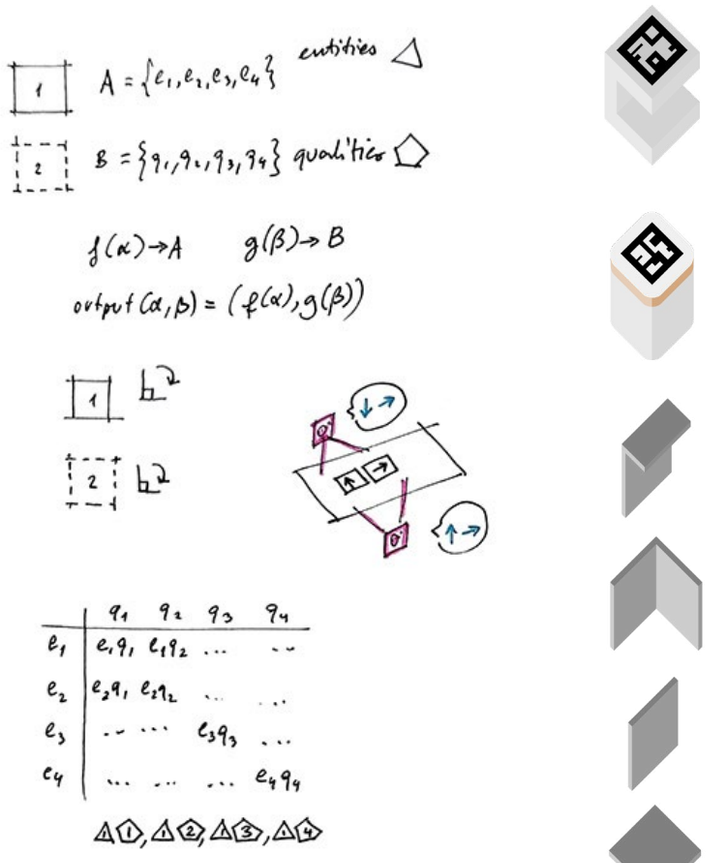




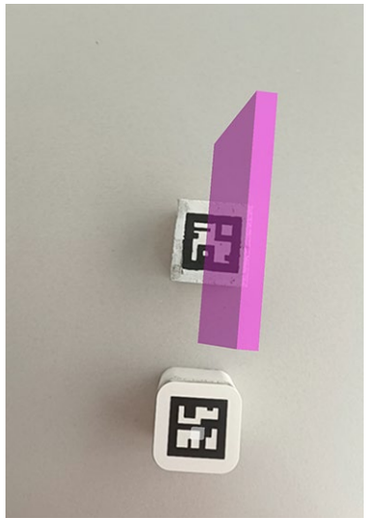


Colour Swap

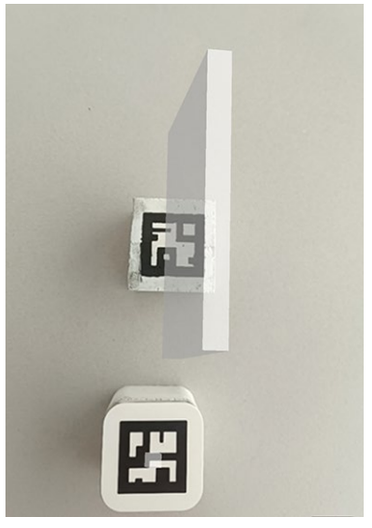
The Colour Swap module introduces a second layer of interaction by allowing one token to influence the visual properties of another overlay. A second marker operates as a switch capable of modifying the colour state of the geometry triggered by a different token. Through rotational tracking, different colour variations can be associated with the same overlay, introducing alternative readings and semantic interpretations within the augmented environment. In architectural workflows, colour often operates as a communicative device for categorisation, hierarchy, or material indication. Within the Augmented Studio, the module explores how abstract visual coding can become part of a participatory and interactive representational language.

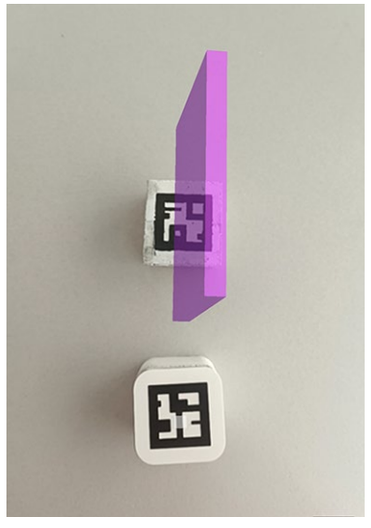


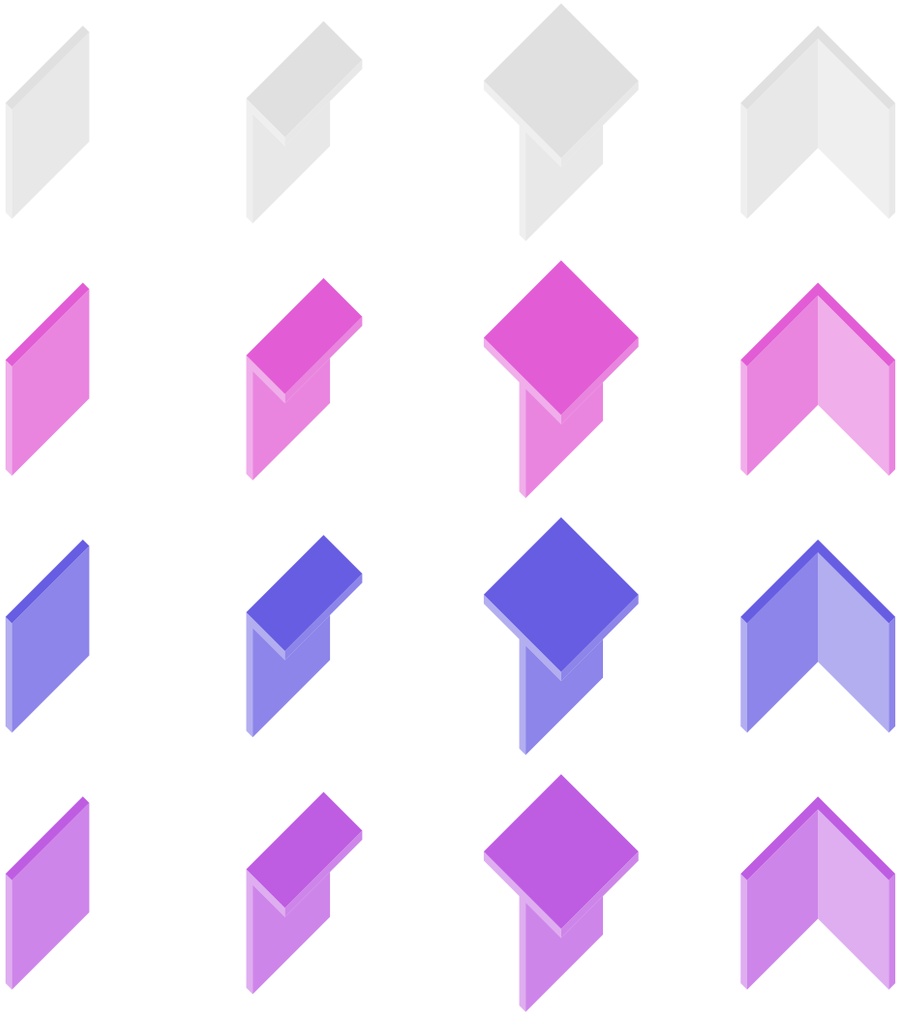






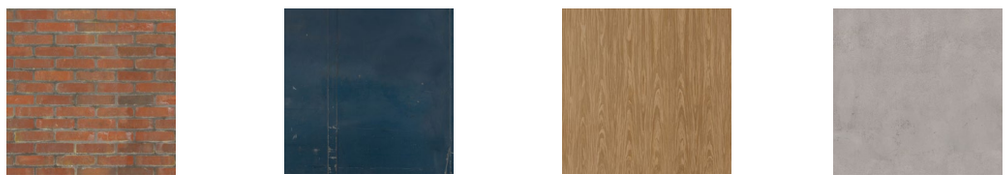






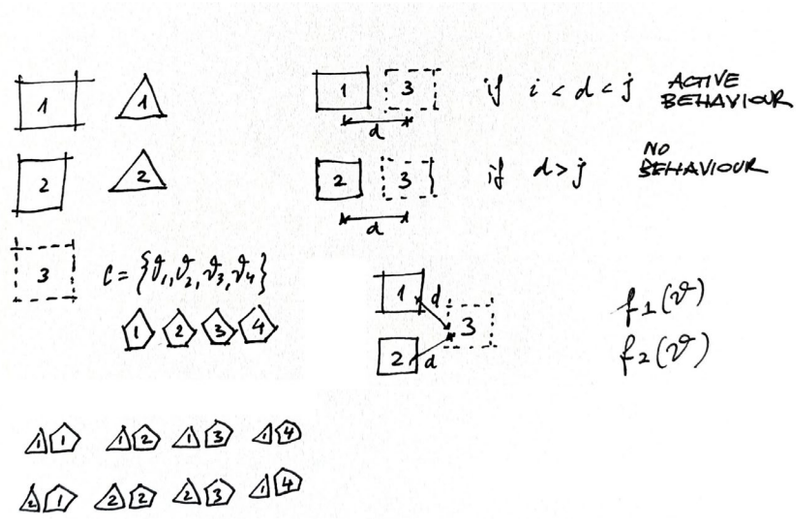
Combinations of the Colour swop

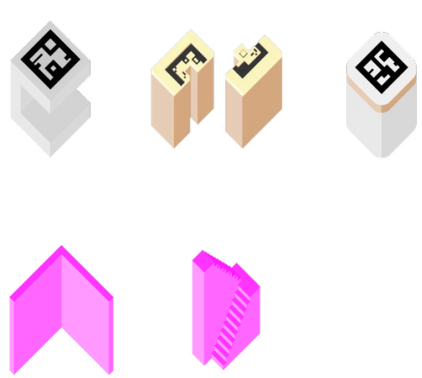
PNG textures

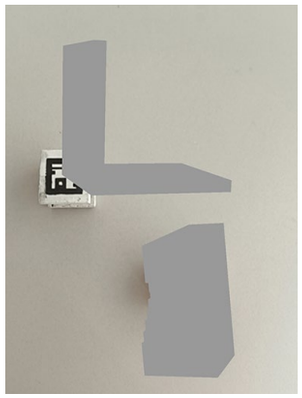


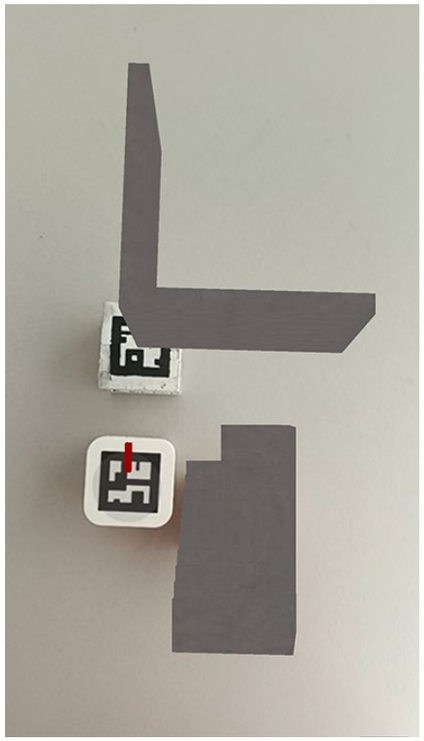
Texture Change

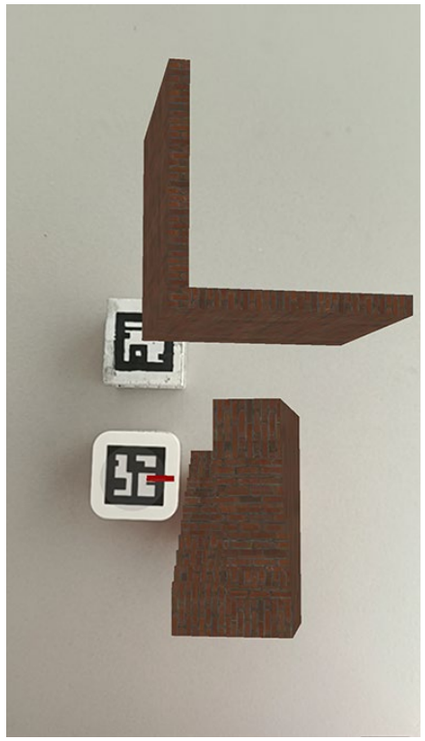
This module expands the logic of the Colour Swap by introducing graphic textures and material qualities into the overlays. Instead of relying on simple colour values, image-based textures are mapped onto the digital geometries, allowing the overlays to communicate additional spatial and material information. The interaction is based on proximity and rotational logic between tokens. A secondary “switch” token modifies the appearance of the primary overlay only when brought within a defined interaction range. Through physical manipulation, users can therefore alter the material expression of the augmented geometry while maintaining the same spatial configuration. The module investigates how material representation and tactile association can enrich the communicative dimension of the hybrid system.

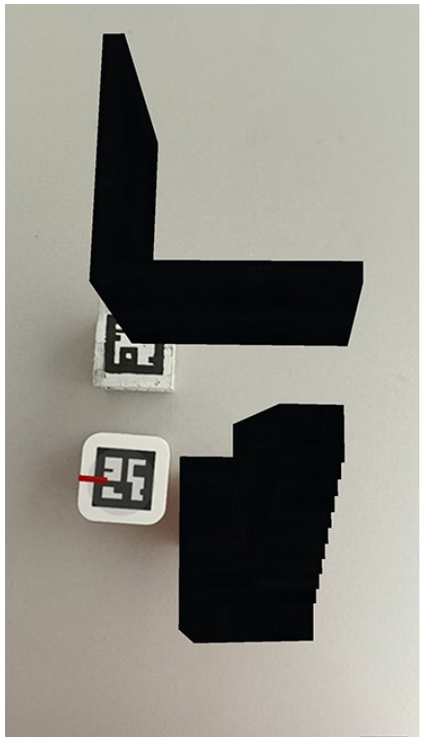


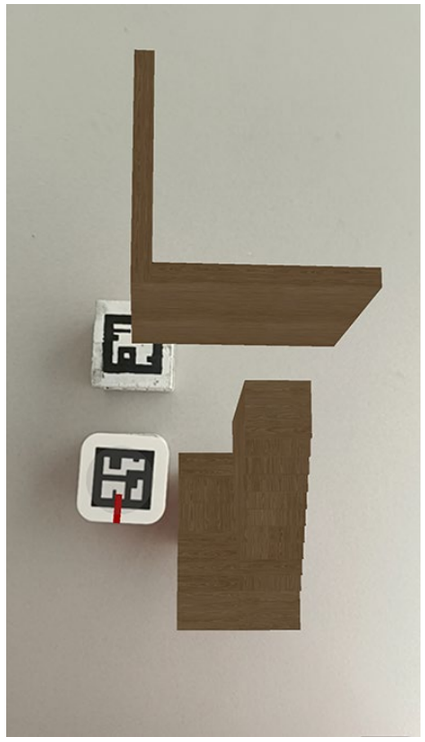






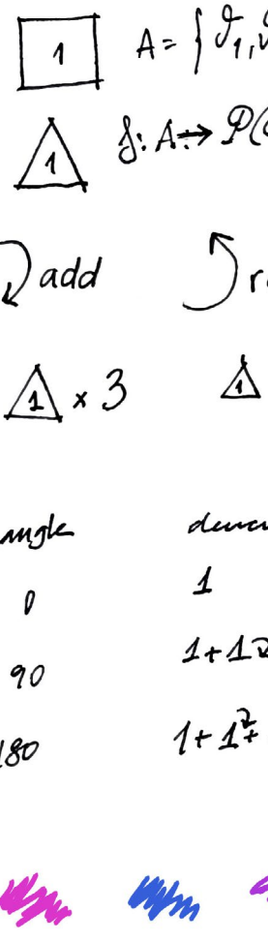


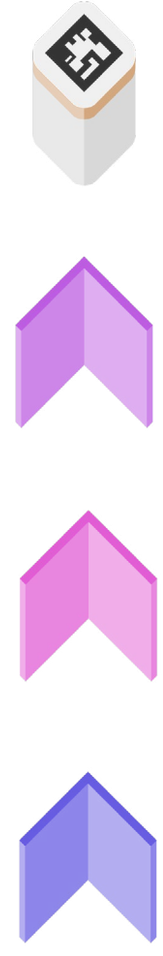


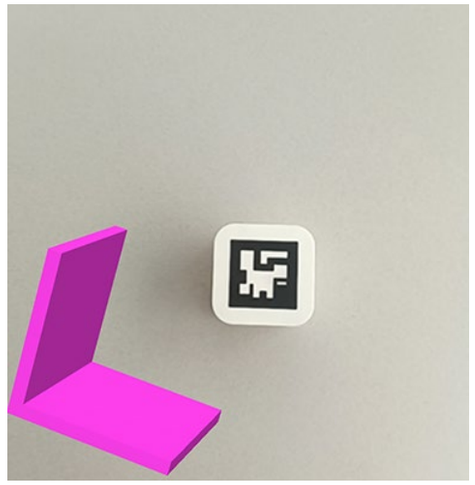


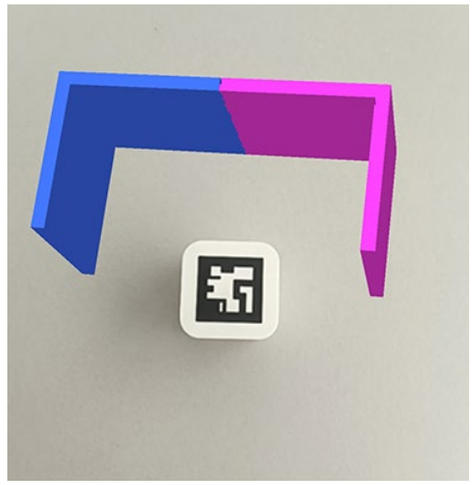
The Room

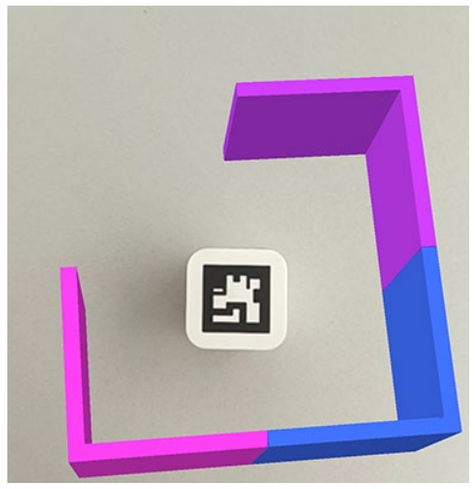
The Room module explores the possibility of constructing spatial configurations through the sequential manipulation of a single token. Instead of triggering isolated overlays, the system progressively composes a larger augmented environment based on the orientation of the marker. Different rotational states generate complementary spatial elements that together define the boundaries of a virtual room. Through repeated manipulation, users can incrementally assemble spatial arrangements and explore compositional relationships within the augmented environment. This module introduces a more architectural mode of interaction, where spatial definition emerges through accumulation, orientation, and modular assembly rather than isolated object placement.





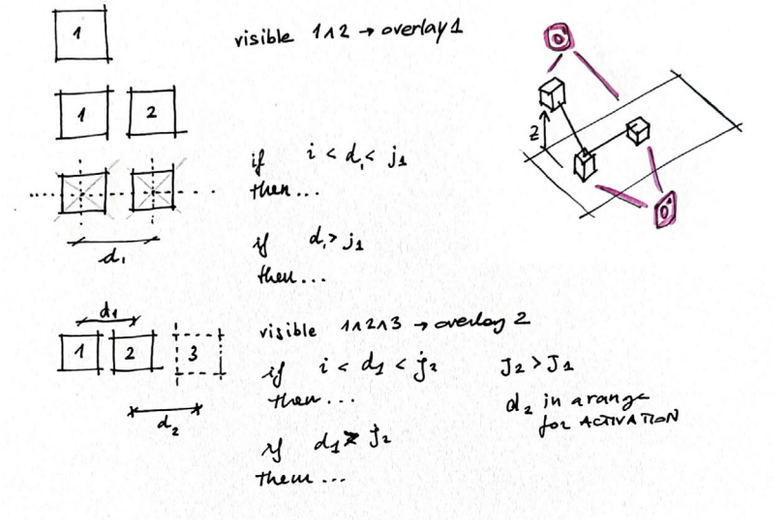




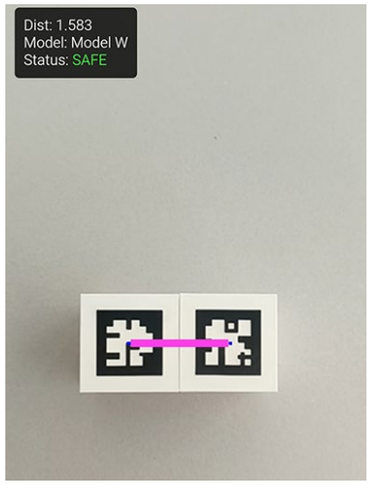


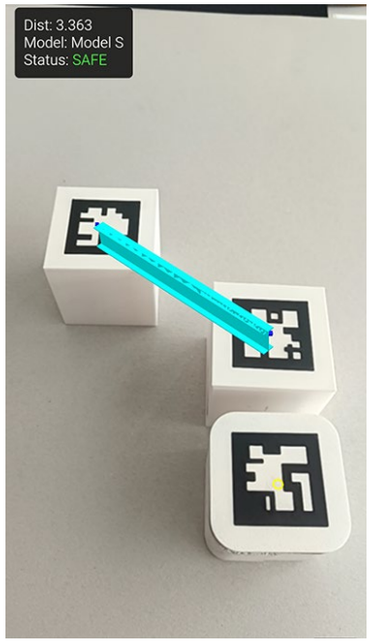
<u>The Beam</u>

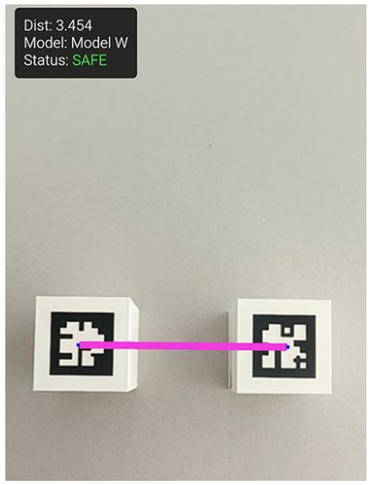
The Beam module investigates dynamic relationships between physical distance and digital transformation. Unlike the previous modules, where overlays primarily respond to discrete states or substitutions, this setup links the geometry of the digital overlay directly to the movement of physical tokens in space.When two markers are detected simultaneously, the overlay stretches and adapts according to the relative distance between the tokens, simulating structural span and spatial extension. Additional markers can further modify the behaviour by introducing alternative overlay states and expanded operational ranges. Through this interaction, the system explores how physical manipulation can communicate spatial constraints, proportional logic, and relational data within the augmented environment. The module therefore combines representational flexibility with performative interaction, suggesting possibilities for more responsive and informative hybrid workflows.

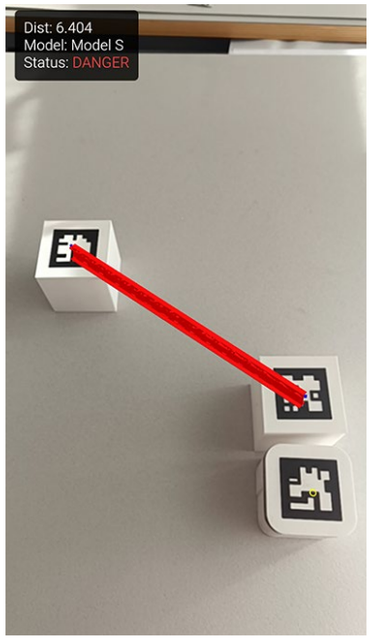


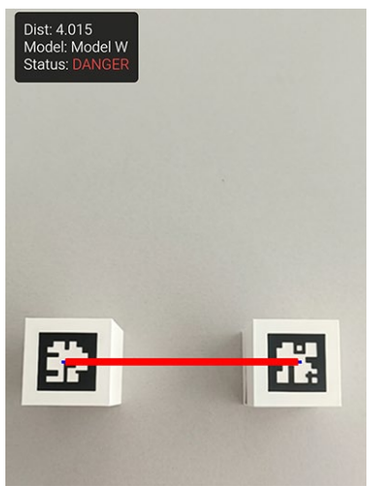












(Left) Overlay_1 triggered by the proximity of the markers and scaled within and above the safe range.

(Right) Overlay_2 .

The development of the behavioural modules shifted the Augmented Studio from a system of static hybrid correspondences towards a more open and relational interaction framework. Rather than operating as fixed representational tools, the modules introduced the possibility for spatial interpretation, transformation, and negotiation through direct physical manipulation and augmented feedback.

Importantly, the resulting interactions remained intentionally simple and accessible. The behaviours were not conceived as finalised applications, but as exploratory structures capable of suggesting alternative relationships between physical artefacts, digital information, and collaborative spatial thinking. In this sense, the value of the system lies less in technical sophistication than in its ability to support flexible and interpretable forms of participation within hybrid design environments.

The chapter has outlined the progressive evolution of the Augmented Studio from early Web-AR experiments into an integrated system of tangible interfaces, representational correspondences, and modular interaction logics. The following chapter extends these observations towards speculative future scenarios and possible modes of deployment within architectural and participatory contexts.

(Left) First marker detected and system searching for secon one.

(Right) Overlay_1 as seen while tilting the camera.







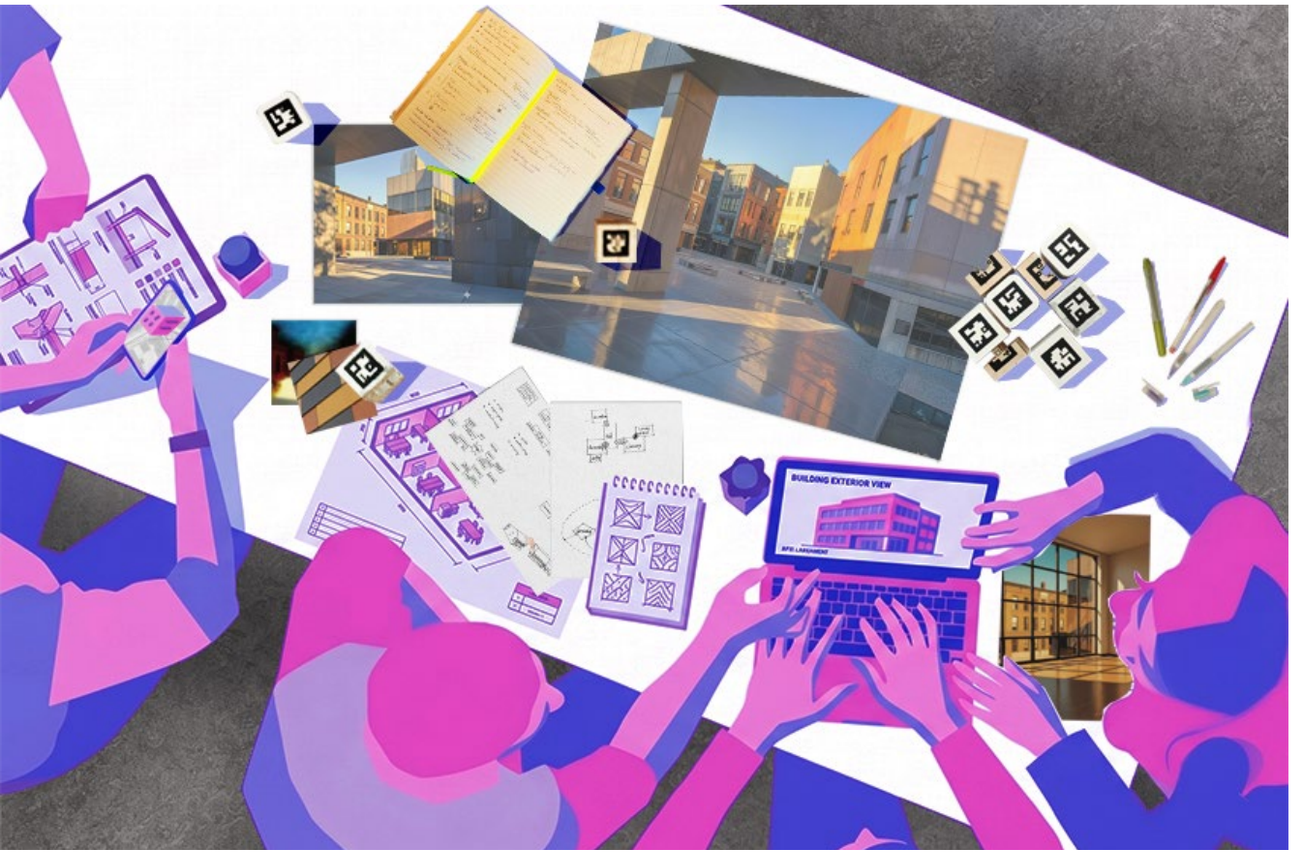
6. Evidence from the future

The following scenarios depict possible situations in which the developed research system could f nd future deployment and further evolution. The accompanying collages operate as a form of "analogue augmentation" of the tangible augmented reality explored throughout the thesis, extending the prototype into speculative yet plausible future contexts.

The various system components developed throughout the course of the research, form physical and digital elements to interaction dynamics, are not final outcomes but rather starting points for inquiry and further exploration. At its current state, Augmented Studio allows for an expansion of the framework with the aim of supporting more interaction logics, updating the AR interface and overall increasing the flexibility and expressive ability of the system. Complexifying the framework should not reduce its accessibility, on the contrary it would maintain the immediacy and broaden the range of communication. The behavioural modules represent a raw state before the transition into larger, everyday life architectural or urban scenarios. They are presented in the following sections.

6.1 Participatory redevelopment scenarios

The first deployment scenario directly draws from the chosen context of intervention: participatory design. The following section illustrates the possible application of Augmented Studio within an urban redevelopment project where habitants of a neighbourhood could contribute to the redesign of their everyday space. In this case, the system would help articulate three phases of work, summarised in the expression “from site to site”. Fostering the integration of more traditional workflows with the technology-enhanced ones, the first phases of informative meetings and charrettes would see the use of AS as a tool for playful rediscovery of one’s surroundings. Similarly to what happens in augmented interpretation panels for tourism, placing QR codes strategically on the site of intervention could help create familiarity, curiosity and confidence in the participants before moving to more design-oriented activities.



The second phase would develop via workshops taking place at the tabletop scale discussed in the system development chapter. Here the physical tokens and augmented reality become the medium of discussion for the group. Here, the use of numerous tangible elements becomes particularly relevant. While similar outcomes could be achieved through a single, physical and reprogrammable object associated with a Web, graphical interface, distributing functions across multiple tactile components would allow interaction to become visible, embodied and negotiated between more participants. Manipulation of the token set would be part of the discussion itself and a result of shared effort, rather than taps on a “Menu”.

In general workshops require guidance, but the suggested workflow would tolerate a looser approach in favour of self-built knowledge and step-by-step understanding. Nevertheless, guidance could be provided in the shape of a catalogue of the available elements and possibilities, some indication of meaning and even some ready-made workflows to familiarize with the system. Rather than prescribing solutions, this material would be in support of exploration and discovery. The physical characteristics of the tokens could also evolve according to the specific context of the workshop. While the current prototype employs generic interaction elements in order to investigate behavioural principles, future implementations might adopt more architecturally meaningful forms (2D and 3D) associated with vegetation, circulation, housing, public amenities, or other project-specific themes.

The last step of the “site to site” scenario would be stepping back into the project location and have the opportunity of visualising, adjust and feel what was produced during the workshops. At this stage the augmented layers would complement the existing environment, making the collective vision more concrete. Upscaling from the tabletop to the urban dimension could require an adjustment in the tracking technology, for the urban scenario would become the trigger for the digital overlays. In this context, the system is not intended to replace professional expertise: architects and planners would remain responsible for evaluating feasibility, synthesizing contributions, and translating outcomes into formal design proposals. The value of the augmented participatory design would therefore lie not in producing complete solutions, but in supporting richer and relatable forms of dialogue and negotiation during the early stages of the design process.



6.2 Educational environments

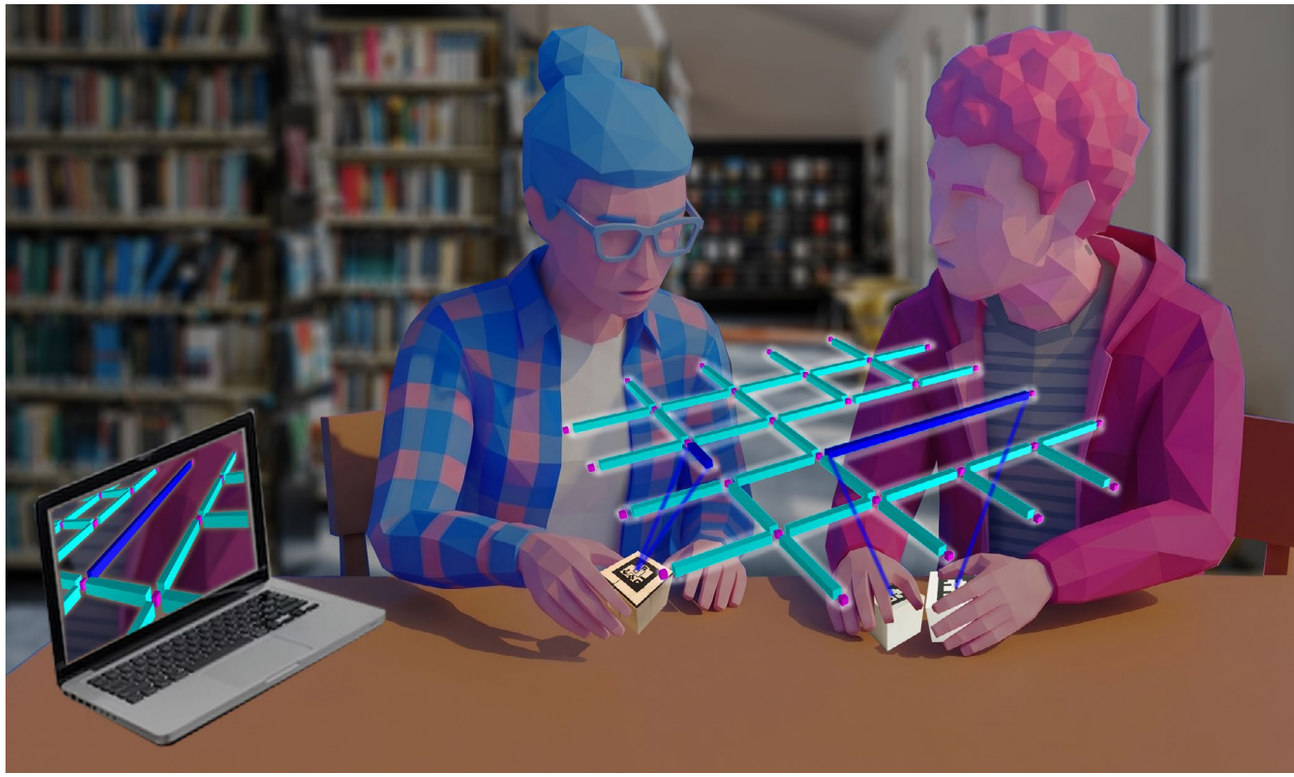
If appropriately refined, a system such as Augmented Studio could support the delivery of more technical ideas. Educational environments could represent a great opportunity in this direction because they are incubators for the skills of future professionals. Compared to more traditional workflows where simulation and modification occur as separate operations, the hybrid interface encourages continuous experimentation through direct manipulation, shifting the way of thinking and designing. This shift is also implicitly toward the more and more common computational mindset: to move beyond simply using the interface, the operative side of a system like AS encourages understanding of scripting and complete workflows that lead to custom solutions to localised problems. In this sense, the Augmented Studio could operate simultaneously as a design tool, a learning environment, and an introduction to hybrid computational workflows.



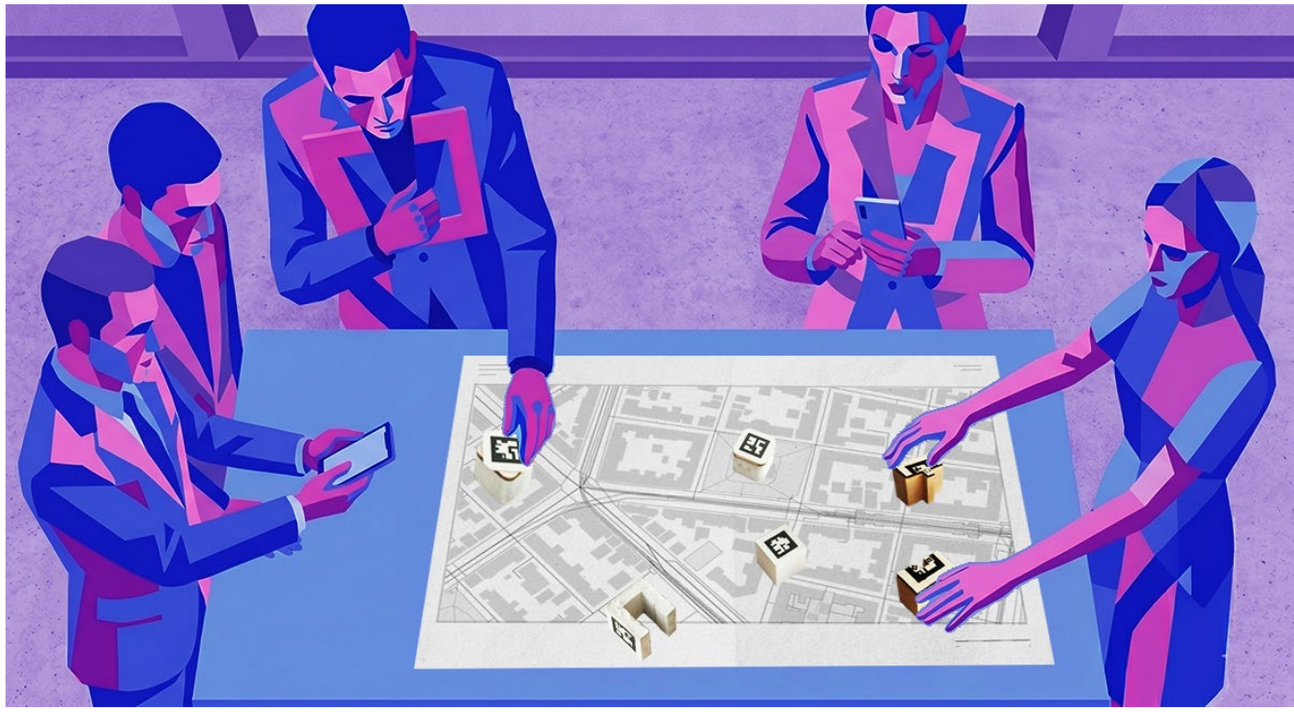
6.3 Distributed architectural practice

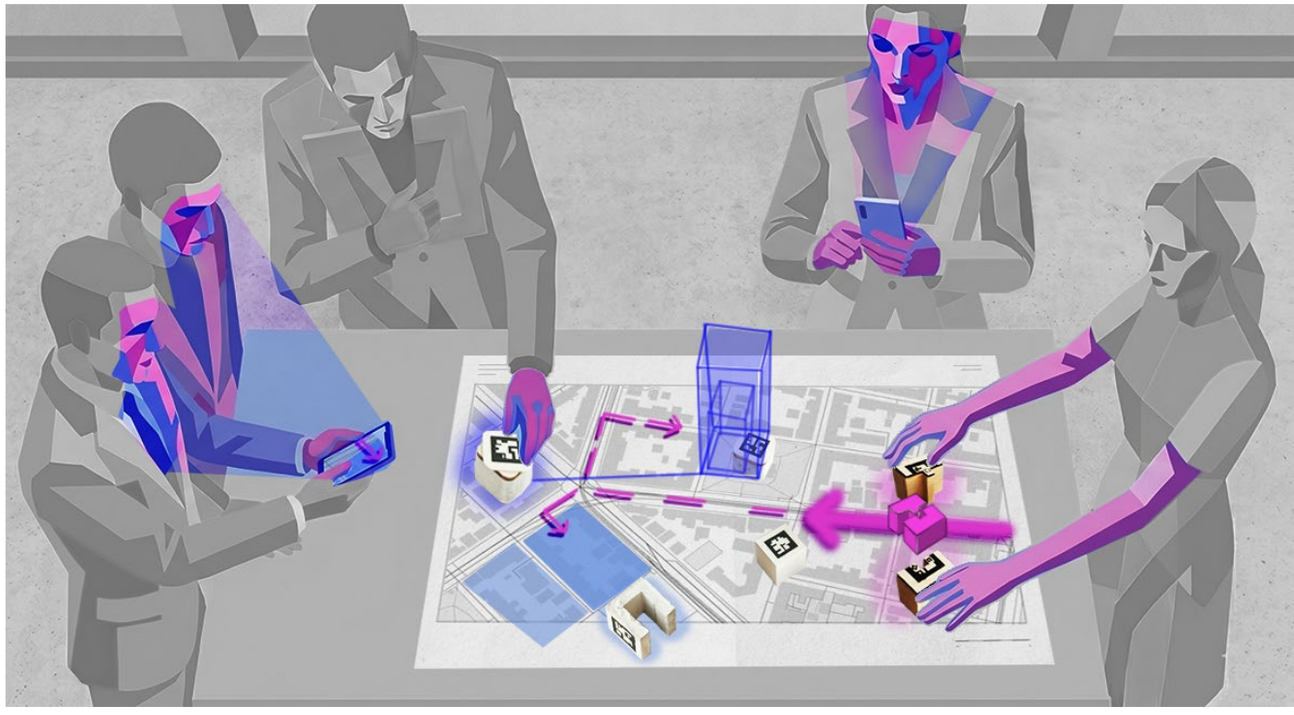
The final scenario concerns the architectural studio itself, where the hybrid system could function both as a design instrument and as a boundary object between architects, collaborators, and clients. In this context, the performance and relational complexity of the behavioural modules could support more dynamic discussions around spatial proposals and project development.

The quality that emerges is this case, is the implicit geography-agnostic character of the system. The stakeholders of a project could manipulate the same representational system, visualise modifications through their own devices, and maintain an ongoing dialogue around the project through affordable and accessible equipment. Moreover, in the practice of architecture, the outcomes of the interactions within the researched system could directly inform collaborative Web CAD platforms where proposals are further refined, documented, and developed, thus translating the general approach also to more executive phases of the design process. This continuity becomes particularly relevant when compared to the fragmented outcomes of more traditional communication dynamics. Rather than operating as isolated systems, hybrid interfaces and Web CAD platforms could therefore form complementary layers within a broader design ecosystem.



The scenarios presented throughout this chapter are not predictions of a singular technological future, but speculative directions emerging from the development of the Augmented Studio framework. Together, they suggest how accessible hybrid interaction systems based on tangible interfaces and Web-AR are relevant within architectural practice. By extending the prototype beyond its immediate technical implementation, the chapter positions the research not as a finished tool, but as an evolving framework for future experimentation and collective spatial communication.







7. Discussion

This chapter presents a critical discussion of the research outcomes, reflecting on the development process, the role of interaction and participation within the framework, its contribution to architectural communication, and potential directions for future development.

7.1 Reflections on prototype development

The discussion of the project draws from the observations made during the process. Several aspects proved successful throughout development. First, the browser-based workflow built on GitHub repositories, Web AR and recurrent code structures remained reliable during both work and demonstration, reinforcing the goal of accessible and easily deployable hybrid interaction systems and the refusal to adopt specialised hardware and proprietary infrastructure. Overall, the project rose interest and engagement both in the informal testing environment and the academic working place, suggesting that the project's features can be solid ground for research. Moreover, the progressive positioning of the project within the participatory architectural design landscape provides a clearer purpose and opens the technical experimentation to broader discussion. The iterative prototype choice, constantly looping and intertwining digital and physical worlds, proved to be the right direction for the process: every step was better than the previous one, and informed the iteration of the following. This is evident in the development of the tangible tokens where the constant rework led to objects with ameliorated affordance, aesthetic and haptic characteristics (e.g. the switch token).

If the tangible parts were easier to control and generated satisfactory outcomes, their digital counterpart were often a source of disappointment. While the project successfully demonstrates several interaction principles, a number of technical and conceptual limitation remain unresolved. The most evident is the stability and reliability if the camera tracking. Even though they don't represent major errors, during a prolonged use of the system, glitches and other inconsistencies can undermine the overall experience. Paired to this, certain interaction logics reveal ambiguities. For example, the rotational behaviours were based on the camera view rather than on absolute spatial reference (tested but unsuccessful), which means that the result of the manipulation of a physical object can be mocked with re-orientation of the camera missing the point of the hybrid system. Digital and physical manage to work together, but the tension between them is not to be overlooked. The first phases of research tried to balance the meaning of the tangible interface and the performance of the augmented layers, but as the process evolved the focus shifted towards implementations of modular behaviours. Broadly, time constraints required choosing between refining the tangible language, expanding interaction possibilities, or improving technical features. Completeness of the system's structure and coherent interaction framework was prioritized over exhaustive refinement of the single element, allowing for a fair evaluation of the work in the frame of participation.

Although this completeness is only relative to the current state of the system and would require substantial refinement before deployment in a real-world scenario, a summative evaluation can nevertheless be made based on the knowledge generated through the process. Beyond the technical solutions developed, the research shows the ability of a hybrid interaction system to support communication, understanding and ideation within architectural practice using simple and raw features. The value of the system is found not only in the visualisation of spatial information, but most importantly in the relations between the involved elements. In fact, some unexpected angles emerged during the process. User engagement appeared to rely more on the physical qualities of the tangible elements than on the visual properties of the marker language itself, pointing to the benefits of embodied interaction in a much stronger way than anticipated. This observation also reframes the tension between physical and digital by translating it between visible and tangible. In a similar way, the complexity of the system requiring collaboration across fields of knowledge to achieve participation becomes a distinctive quality of the work method suggested in the research. Regardless the need for further discussion, these ambiguities and considerations suggest that the contribution of the hybrid systems may reside less in their ability to display proposals and more in their capacity to structure exploration, confrontation and negotiation around them.

7.2 From refinement to interaction

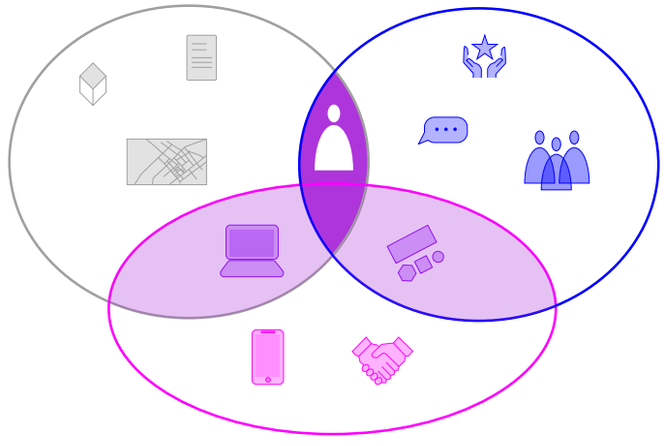
As mentioned, during the development of the system it was necessary to shift focus between the various pieces that compose the outcome. But this shift also revealed how the project changed. At the beginning the emphasis was put on the semiotic aspect thanks to which the relation between markers, tokens and AR overlays delivered meaning. This proved to be a good starting point but created a juncture: either keep working on the single elements and perfection them or put them into dialogue and relation. The first option was carried out to some extent through the refinement of the physical tokens' dimensions, marker integration and affordance. But overall, the refinement was not supposed to have and end in itself, mainly because it would create a series of nice objects for contemplation and not much more. Instead, preferring to work on the relation between physical parts and their augmentation in the digital world naturally brought up the right choices of refinement and added value to the system by turning it into a deployable toolset. As manipulation, aggregation, proximity, and other relational behaviours were studied the taxonomy became necessary for organising and understanding the growing complexity of the system. Introduced not to categorise individual artefacts, but to define the relation between action and digital response, the taxonomy symbolizes the change of perspective that prioritises enabling users to do something over individual object design. In this sense, interaction took over representation and meaning started emerging from action rather than through static association of signifier and signified. By remaining complementary the explorative tangible interface and the informative digital overlay evolved form representational device to interaction framework in support of discussion and experimentation.

7.3 Participation as mediation

The transition from representation to interaction is the cornerstone for discussing the system within PAD processes. Over the course of the research participation stood out as the most viable context for the practical application of the research. The technical framework did not loose its importance, but rather it became stronger and justified by its ability to assist human interaction. In fact, this argument is directly connected to the original problem behind this research: the communication gap created by increasingly complex, black-boxed digital infrastructures and technical workflows used in architecture practice. Mediating between professionals and non-professional actors remain a challenge even though participation is a recurrent component of design nowadays. In this frame, the project is not replacing participatory methods but is added as a support of communication and understanding.

A similar, although more layered, role is played by the architect. The professional figure is not substituted, authorship is not transferred and participants do not take responsibility of the executive design aspects. Nevertheless, for the first phases of a design the architect would have to step “back and out”, meaning that its first job is designing the hybrid ecosystem of interaction and establish the conditions for the participatory process to unfold. Then the focus would shift to assisting, collecting, interpreting and evaluating results. Participation is here defined not as the elimination of expertise, but as the openness to different forms of knowledge within a shared environment.

Diagram showing how the intersection of professional and public word becomes a fertile ground for interaction thank to the mediation of the architect and the hybrid system.



In fact, the system does not aim to generate design proposals and, even if it can be used in brainstorming and sketching raw ideas, its value emerges in structuring dialogue. Participants are not limited to expressing preference or reacting to something ready-made. On the contrary, the toolset allows active engagement and explication of situated knowledge so that the augmented space is a medium between pure abstraction and more formalised architectural design. As a consequence, the use of web-based technology finds its justification and relevance in the participatory context, while the contribution of the project extends beyond technical prototyping into accessible and meaningful collaborative design methods.

7.4 Contribution through intergation

When defining the contribution of this research it is evident that it is not about inventing a new XR technology or discovering tangible interfaces. The chosen components are not entirely new, nor is the aim to propose an alternative to existing infrastructure. Web AR, marker-based tracking, browser-based workflows, and tangible interaction have independent precedents in research and professional contexts. Instead, the research’s contribution resides in the integration of these components into one coherent framework positioned within participatory architectural design, where accessibility is the link between the involved parts. Combined, the elements converge into an alternative method of work in support of communication and understanding.

The integrative and collaborative approach is reflected in the technical structure of the project. Displayed in the table on the text pages are the various open-source technologies and libraries used to create the AR environments and behavioural modules, as well as the design and manufacture tools used for manufacturing of the physical objects.

Overall, the developments of the process showed how the dialogue between frameworks can generate new opportunities for architectural experimentation, and at the same time the iterative prototyping reshaped the theoretical positions through practical testing. Consequently, the result of the research is not a finished platform or a complete participatory methodology. It rather highlights the potential of underused, existing technologies in re-emerging co-design contexts.

Tool	Phase of use	Role in the workflow
Rhino	*Design and modelling*	*Development of digital geometries, AR assets, and physical token prototypes.*
Adobe Illustator	*Graphic design*	*Design of markers, visual identity, diagrams, and communication material.*
Bambu Studio + Bambu Lab 3D printer	*Digital fabrication*	*Production and iterative refinement of tangible tokens and physical interaction artefacts.*
Lightburn + Falcon2pro laser cutter	*Digital fabrication*	*Manufacturing and engraving of markers and supporting physical components.*
MindAR	*Web AR tracking library*	*Marker detection, image tracking, and positioning of augmented content.*
A-frame	*Web XR framework*	*Construction of the browser-based augmented environment and management of AR entities.*
JavaScript	*Programming language*	*Development of behavioural modules, interaction logic, proximity calculations, rotation tracking, and system responses.*
GitHub Repositories + GitHub Pages	*Version control and deployment*	*Repository management, iterative development, and browser-based distribution of the prototype.*
Mobile Browser + camera device	*Access platform*	*User interface for experiencing and interacting with the system without dedicated hardware.*
Behavioural taxonomy	*Design framework*	*Organisation, evaluation, and categorisation of interactions throughout development.*

7.5 Future development and open questions

The viability of the researched framework is demonstrated through the prototype, but several questions emerge from it and remain open. Inquiry on deployment, scalability and integration of the method in everyday practice do not necessarily represent shortcomings but rather new directions that could not be explored within the timeframe of this work.

As suggested in the previous chapter, the participatory scenarios generate results that are often difficult to translate into formal practice. The tangible manipulation, discussion between participants, reconfigurations of the augmented layers can elude the knowledge transfer. To resolve this ambiguity future developments could investigate ways of recording, saving, annotating and documenting what emerges from the use of the system. It could be done both on the technical level (the code) or on practical (pictures, screenshots). The important thing is that the material remains accessible for the architect to be able to rework it and for the user to follow-up on the process. In this sense, emerging browser-based CAD ecosystems and collaborative design platforms can become the synthesis medium for the design phases following the participatory workshops.

A second reflection points to the nature of the tangible interface itself. During the research, tokens shifted from tangible signifiers to abstract interaction devices defined by their relationship with digital counterparts. In this way the toolset is reusable because it can be reprogrammed, but external feedback suggested that different scenarios could require ad hoc solutions with more explicit connections between the artefacts and their architectural interpretation. Shifting to scenario-specific tokens whose shape, materiality or symbolism effectively communicate their role within the participatory process should be explored in following iterations. The tension between flexibility and specificity is an open design question, and it does not ask for a unique answer.

Lastly, the doubts about the efficacy of the tabletop setup are justified by the fact that architecture exists at many different scales, and the 1:1 scale is the aim of every project. The tabletop is not necessarily the only possible setting, even though it supports the discussion and collaboration topics addressed by the research. But the tabletop can be seen as a compromise needed to build the sharing understanding, so that the translation into larger environment or urban spaces can happen smoothly. It is also true that these contexts could provide with novel insights of the participatory processes. Thus, future developments could reimagine the technical framework to adapt it to larger scenes (e.g. a picture of a whole building could become the marker for triggering the AR). An outcome of this sort would push the research towards the immersive XR technologies, which are not the object of this thesis.

Together, these reflections strengthen the openness of the research, restating its value not only in the prototype but in the inquiry role it plays within co-design, architectural practice and the future of hybrid tool systems.

8. Conclusion

This thesis investigated the potential of hybrid interaction systems as mediators between professional expertise and participation processes. Moving from the communication gap widened by increasingly complex digital workflows, the work explored how tangible interaction and web-based augmented reality can cooperate in support of professional-to-laypeople dialogue. Applying a Research through Design methodology based on iterative prototyping, the work developed into “Augmented Studio”, a hybrid framework combining tangible artefacts, digital overlays, and behavioural interaction.

The process led to multiple findings. The main one concerns the general focus of the work: while the early phases were centred around markers, signifiers, and visual associations, later steps revealed that the relationship established between the physical actions and digital responses is where the value of the system really belongs to. Meaning became more and more related to manipulation and negotiation than to direct representation.

This transition was reflected in the elaboration of the behavioural modules and the taxonomy re-orienting the project towards a proper interaction framework and away for a mere visualisation device. In fact, it is also true that various successful aspects of the project were not the most visually refined. The conducted informal test showed that touch comes naturally before detailed vision, so direct manipulation resulted more meaningful than observation in this case. Affordance, tactility and embodied interaction played a strong role and pointed out that the hybrid system balance between physical

and digital is delicate and easily disrupted. Given the increasing digitalisation of design environments, the research tries to reinforce the relevance of the tactile interaction.

Another major finding concerns the role of participation. Without overtaking the technical infrastructure, participatory design emerged as the optimal context for the deployment of the researched method. This allowed the technical frame to gain relevance and be essential to the dialogue across different levels of expertise. The hybrid environment is not meant to replace professional knowledge or distribute authorship. It humbly provides a shared space to express and discuss situated knowledge, while reinforcing the role of the architect as a designer of the interaction environment, facilitator and translator.

This redefinition of the architect's role directly feeds the complexity argument presented at the beginning of this work. In this regard, the research is open and embraces complexity by taking it as a challenge and trying to resolve it into a productive working condition (i.e. the combination and relation of the various parts of the system). Rather than reducing complexity, the project explored a way of making it accessible and negotiable.

The main contribution of this research stands in suggesting how existing technologies, design methods and theories can be combined into a harmonious hybrid interaction system for architectural communication. The objective was establishing the common framework for Web AR, tangible interaction and participatory design prnciples. Iterative prototyping brought together theories of affordance, semiotics and co-design through their constant reshaping and testing. The final result is not only a prototype but a method and a position regarding the role of accessible and open technology in the architectural field of the future. The questions that emerged around Augmented Studio point towards further research and development rather than to the limits of the concept itself.

Ultimately, the research states that the importance of such hybrid systems unarguably lies in their ability to display information, but also in their ability to construct communication around it. By creating ecosystems where physical actions, digital feedback, and collaborative discussion consciously intersect, hybrid interaction systems trigger the rethinking of how architectural knowledge is shared, negotiated, and collectively explored.

9. Acknowledgements

Personal acknowledgements

I would like to acknowledge my supervisor, Pablo Miranda, and my examiner David Andréen, for their role in the supervision and evaluation of this thesis.

Thanks to my collegues for showing interest about the project during the hours in the studio.

Statement on the Use of Artificial Intelligence

AI tools, including Google Gemini and ChatGPT, were used during the development of this thesis as support for language refinement, formatting, code scripting assistance, image editing, and critical reflection throughout the writing and design process. However AI was not used for producing original ideas: foundational concepts, arguements and conclusions are original work of the author.

Unless otherwise indicated through citations and references, all graphic material, diagrams, visualisations, photographs, and design outcomes included in the thesis were produced by the author.

Bibliography

Abhari, M., Abhari, K., Drinkwine, M., & Sloan, J. (2021). E*xtended Reality (XR) Applications in Architectural Practice: Towards a Development Framework.* In C. Stephanidis, M. Kurosu, J. Y. C. Chen, G. Fragomeni, N. Streitz, S. Konomi, H. Degen, & S. Ntoa (Eds), HCI International 2021—Late Breaking Papers: Multimodality, eXtended Reality, and Artificial Intelligence (Vol. 13095, pp. 185–196). Springer International Publishing. https://doi.org/10.1007/978-3-030-90963-5_15

Arnstein, S. R. (1969). *A Ladder Of Citizen Participation.* Journal of the American Institute of Planners, 35(4), 216–224. https://doi.org/10.1080/01944366908977225

Billinghurst, M., Kato, H., & Poupyrev, I. (2008). *Tangible augmented reality.* ACM SIGGRAPH ASIA 2008 Courses on - SIGGRAPH Asia '08, 1–10. https://doi.org/10.1145/1508044.1508051

Blundell-Jones, P., Petrescu, D., & Till, J. (Eds). (2012). *Architecture and participation.* Routledge. https://doi.org/10.4324/9780203022863

Bolter, J. D., & Grusin, R. (2002). *Remediation: Understanding new media* (5. pr). MIT Press.

Eco, U. (1980). *Function and sign: the semiotic of architecture*. In Broadbent, G., Bunt, R., & Jencks, G. (1980). Signs, symbols and architecture. J. Wiley and sons.

Ehn, P. (1992). *Setting the Stage for Design as Action - artifacts for participatory de¬sign in theory and practice.* Nordisk arkitekturforskning, 5(4), 49-59.

Frayling, C. (1993). *Research in art and design.* Royal College of Art Research Papers, 1(1).

Gaete Cruz, M., Ersoy, A., Czischke, D., & Van Bueren, E. (2022). *A Framework for Co-Design Processes and Visual Collaborative*

Methods: An Action Research Through Design in Chile. Urban Planning, 7(3). https://doi.org/10.17645/up.v7i3.5349

Gaver, W. (2012). *What should we expect from research through design?* Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, 937–946. https://doi.org/10.1145/2207676.2208538

Gibson, J. J. (1986). *The ecological approach to visual perception.* Erlbaum.

Hu, W., Kahlon, Y., Nakatani, M., & Fujii, H. (2025a). CI-3. *Rethinking Participatory Architectural Design: Harnessing Immersive Environments & AI for Facilitating Designer-Stakeholder Collaboration* (p. 4704465 Bytes). HKU Data Repository. https://doi.org/10.25442/HKU.29345678

Hu, W., Kahlon, Y., Nakatani, M., & Fujii, H. (2025b). *Empowering End-Users in Spatial Design Tasks: An empirical study of co-design in extended reality.* 347–356. https://doi.org/10.52842/conf.caadria.2025.4.347

Ishii, H., & Ullmer, B. (1997). *Tangible bits: Towards seamless interfaces between people, bits and atoms*. Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems, 234–241. https://doi.org/10.1145/258549.258715

Latour, B. (1987). *Science in action: How to follow scientists and engineers through society*. Harvard University press.

Maeda, J. (2006). *The Laws of Simplicity.* MIT Press.

Monteiro, K., Vatsal, R., Chulpongsatorn, N., Parnami, A., & Suzuki, R. (2023). *Teachable Reality: Prototyping Tangible Augmented Reality with Everyday Objects by Leveraging Interactive Machine Teaching.* Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, 1–15. https://doi.org/10.1145/3544548.3581449

Norman, D. A. (2013). *The design of everyday things* (Revised and expanded ed). Basic Books.

Peirce, C. S. (1931–1958). *Collected papers of Charles Sanders Peirce* (C. Hartshorne, P. Weiss, & A. W. Burks, Eds.; Vols. 1–8). Harvard University Press.

Reaver, K. (2022). *Mixed Reality in Multiuser Participatory Design: Case Study of the Design of the 2022 Nordic Pavilion Exhibition at the Venice Biennale.* Buildings, 12(11), 1920. https://doi.org/10.3390/buildings12111920

Redström, J. (2021). *Research through and through design.* Artifact, 8(1), 16.1-16.19. https://doi.org/10.1386/art_00016_1

Rojtberg, P., & Olwal, A. (n.d.). *Tangible Interfaces using Handheld Augmented Reality.*

Safin, S., & Dorta, T. (2020). *Unfolding Laypersons Creativity Through Social VR - A case study.* 355–364. https://doi.org/10.52842/conf.ecaade.2020.1.355

Sanders, E. B.-N., & Stappers, P. J. (2008). *Co-creation and the new landscapes of design.* CoDesign, 4(1), 5–18. https://doi.org/10.1080/15710880701875068

Sanoff, H. (2000). *Community Participation Methods in Design and Planning.* 10.1016/S0169-2046(00)00063-3.

Schön, D. A. (2017). *The Reflective Practitioner* (0 edn). Routledge. https://doi.org/10.4324/9781315237473

Sikiaridi, E., & Vogelaar, F. (2012, November 5). *Hybrid Space Lab and Networked Participatory Design Systems.* World-Architects. https://www.world-architects.com/en/architecture-news/insight/hybrid-space-lab-and-networked-participatory-design-systems

Söderlund, C., & Evans, P. (2022). *Co-design in immersive VR: Something old, something new, something else….* In T. Oksala, T. Orel, A. Mutanen, M. Friman, J. Lamberg, & M. Hintsa (Eds.), Craft, technology and design (pp. 216–235). The Finnish Society for Practice Based Inquiry (PraBa) & Häme University of Applied Sciences. https://www.diva-portal.org/smash/record.jsf?pid=diva2:1654630

Star, S. L., & Griesemer, J. R. (2016). *Institutional Ecology, "Translations," and Boundary Objects: Amateurs and Professionals in Berkeley's Museum of Vertebrate Zoology*, 1907–1939. In G. C. Bowker, S. Timmermans, A. E. Clarke, & E. Balka (Eds), Boundary Objects and Beyond (pp. 171–200). The MIT Press. https://doi.org/10.7551/mitpress/10113.003.0011

Systèmes Hybridlab. (2021). Hyve-3D. https://www.hyve3d.com/

Whyte, J. (2002). *Virtual reality and the built environment* (1. publ). Architectural Press.

ZHVR Group. (2018, October 20). *Project Correl*. https://www.zhvrgroup.com/project-correl

Software, Frameworks and Online Resources

AR.js Contributors. (n.d.). AR.js [JavaScript library]. https://ar-js-org.github.io/AR.js/

Cabello, R., & Three.js Contributors. (n.d.). Three.js [JavaScript library]. https://threejs.org/

Kim, H. (n.d.). MindAR.js [JavaScript library]. https://github.com/hiukim/mind-ar-js

Mozilla. (n.d.). A-Frame [Web framework]. https://aframe.io/